Automatic GFL/GFM Reference-Mode Switching in a 1-SG / 4-IBR IEEE-14 System:

Model, Small-Signal Spectrum, Switched Time-Domain Formulation, and a Five-Policy Comparison on One Trajectory

(250.000 s **result set)**

Power-flow / stability toolchain — generated report

Abstract

A modified IEEE-14 network is driven through a six-period chronology in which a single synchronous generator (SG) is tripped, the load is stepped, a shunt fault is applied and cleared, a tie line is opened, and the SG is finally reclosed. Four inverter-based resources (IBRs) switch *automatically* between grid-following (GFL) and grid-forming (GFM) reference behaviour under an amplitude–frequency severity index with hysteresis and dwell. This report documents, from the production code, (i) the severity index and its two component sub-indices J_V, J_f, together with the four reference-only diagnostic indices published alongside them; (ii) the sixth-order SG model with an order-by-order derivation and the singular limit that freezes one flux state; (iii) the seventeen-state dual-mode IBR superset with its active projections and current-continuity-gated transfer; (iv) the switched implicit-trapezoidal time-domain formulation, including a proof of the block structure of the stacked right-hand side and the exact mechanism by which an integrated state update is replaced by a discrete transfer at a mode change; (v) the solved power flow, the small-signal spectrum, and the accepted transient response of the 250.000 s run; and (vi) a comparison of *five* control policies — all-GFL locked, one, two and four pinned grid-forming units, and the delivered adaptive supervisor — on the same network, dispatch, event schedule and solver. All equations are those the run executes; all plotted curves are raw accepted samples of that run. No claim of operational readiness is made: convergence of the simulation is not evidence of grid-code compliance.

Contents

1 Introduction

Displacing synchronous machines with converter-interfaced generation removes rotating inertia from the system, and the consequences for frequency stability are by now well characterised: the review in [12] surveys the mechanisms and the control responses, [13] proposes a quantitative inertia-security assessment for exactly this regime, and [14] is one of many synthetic inertia schemes that answer the shortfall from the converter side. What that literature makes clear is that the shortfall is not only a matter of how much inertia is emulated, but of *which converters are willing to form a voltage* when the last machine leaves.

Grid-forming (GFM) control answers that question by making the converter a voltage source with its own angle, the mode whose autonomous operation was characterised early for inverter-based microgrids [9], and grid-following (GFL) control declines it by synchronising to a measured angle through a phase-locked loop. Neither is free. GFM requires active and reactive headroom to be reserved, which is why [15] treats the control mode as a dispatchable grid service with a steady-state cost rather than as a fixed property of a device, and why the capability envelope of a grid-forming inverter has needed explicit characterisation [25]. Under a fault, the two modes behave differently once the current limit binds [24], which is where the comparison stops being a matter of taste.

That cost is what motivates *switching* rather than choosing. Designs that let one converter present either behaviour have been proposed with minimal structural change [16], adapted continuously to the short-circuit ratio [17], made bumpless at the transition [18], or unified into a single grid-following framework that also survives weak grids [19]. In parallel, the transient behaviour of systems that mix machines with both converter types has been analysed directly: [20] for inverter-based generation against a machine, [21] for the damping interaction in a hybrid GFL-GFM pair, [23] for a parallel SG-GFL-GFM system, and—closest to the mechanism this report ends up measuring—[22], which studies what the *switching of* droop-controlled grid-forming converters does to the transient stability of the grid-following converters around them.

What is not settled by that work. Those studies establish that switching helps and that mixed systems have coupled transient behaviour. They do not, between them, answer the question a system operator actually faces on a multi-machine network: given a set of converters that *could* form a voltage, how many should, which ones, and by what route. Two gaps in particular motivate this report. First, a configuration can be admissible in the small-signal sense and still be unreachable from the present state, so a selection rule built on linear screening alone is not sufficient. Second, the number of grid-forming units is usually treated as monotone in benefit, and on the system studied here it is not.

Contributions. This report is a numerical study of one modified IEEE 14-bus system with a single synchronous machine and four converters, driven through a six-period disturbance chronology, and it contributes the following.

1. A complete transcription, from the executing code, of the switched composite model: the severity index that decides the mode (Section 4), the sixth-order machine with its singular-limit flux reduction (Section 5), the seventeen-state dual-mode converter with its current-continuity-gated transfer (Section 6), and the implicit-trapezoidal switched formulation (Section 7).
2. A proof, from stated hypotheses rather than from the implementation, that the composite right-hand side is the direct sum of per-device fields, with a converse showing the hypotheses are necessary and a counterexample drawn from a quantity this system actually computes (Section 7.2).
3. An exhaustive small-signal screening of every candidate grid-forming configuration in the islanded band, which finds only 7 of 15 admissible and the best margin at *two* units rather than four (Section 9).
4. A five-policy comparison on one trajectory — all-GFL locked, one, two and four pinned grid-forming units, and the adaptive supervisor — which shows that promoting at least one converter is a necessary condition for a post-trip trajectory to exist at all, that the four-unit configuration fails when pinned from the trip although it passes the linear screen, and that the adaptive arm reaches that same configuration and survives by staging the transitions through per-step certificates. Those five arms are run and their outcomes are used below where they bear on a claim; the comparison itself is held out of this revision and will be presented separately.

What this report does not claim.

- **No operational readiness.** Convergence of a simulation is not evidence of grid-code compliance, protection coordination or hardware realisability. Transient extrema in what follows are honest excursions of the model under a severe chronology, not operating values or ratings.
- **No selector-against-selector comparison.** The comparison is switching against not switching on *this* resource mix. No claim is made about other selectors, or about systems containing further voltage-forming sources.
- **No multi-machine result.** The repository carries per-machine dynamic data for the reference bus only and the hybrid path refuses several synchronous machines at four fail-closed points, so no numbers for that case appear here and none are assumed in their place.
- **Only S, J_V and J_f decide anything.** J_R, J_P, J_{lock} and J_{SCR} are computed after the trajectory is complete, enter no gate, are classified ASSUMED_DIAGNOSTIC, and must not support a readiness claim. No aggregate is formed from them by design.
- **One linearisation per point.** The spectrum of Section 9 and every damping margin in the admissibility table are single operating-point analyses and do not bound the switched system. The four-unit configuration is admissible by that screen and is still not reachable when pinned from the trip, which is a limit of the linear tool itself and not of the screen's arithmetic.
- **DC-link dynamics are at the resolution limit of the method.** The DC source is non-ideal (Section 6.3) and its mode is faster than the nominal phasor step, so it is resolved only where the adaptive controller reduces the step. A firmer DC source is stiffer still and belongs to an EMT study rather than to this one.
- **The DC circuit responds but cannot act back.** There is no modulation-index constraint tying the commanded AC voltage to the available V_{dc}, so $\partial f_i/\partial x_j = 0$ for every AC state i and both DC coordinates j, and the coupling is one-way. A sagging link therefore never clips an AC command in this model, and no conclusion here may be read as covering DC-limited ride-through.
- **Cross-arm numerical quality is not compared.** Residuals and rejected-step counts are not comparable between trajectories of unequal length, and an arm that ends early is never extended to match the others.

2 Scope, sources and reproduction

The study system is the IEEE-14 bus network with one modelled SG at the reference bus and four modelled IBRs at buses 2, 3, 6 and 8. The whole system is assembled as a single all-KCL composite differential–algebraic system

$$g(x, V) \;=\; Y_{\text{net}}\, V \;-\; I_{\text{bus}}(x, V) \;=\; 0, \tag{1}$$

where Y_{net} is the network admittance, V the stacked complex bus voltages, and I_{bus} the total device injection. The SG and every IBR contribute their own differential state block x and their own injection into (1); no device is delegated to an external solver, model or reference program. Audited linear primitives (`\`, `lu`, `qr`, `eig`) are the only numerical libraries used.

Equation source. Every model equation in Sections 4–7 is transcribed verbatim from the production code that the chronology driver executes. The network, dispatch, event schedule and supervisor gains are fixed by the driver itself, so the whole result set is reproduced by

```
pf_init_paths; run_ieee14_gfm_lock_comparison  % t_end=250, dt=0.05
pf_init_paths; generate_ieee14_switch_evidence
```

The first command produces the five arms; the second draws every figure and writes every table and macro this report inputs. The comparison runner builds *one* base option set, adds the opt-in post-processing overlay described next, and then merges a declared per-arm override; it aborts if the two option structures differ in any field that the arm did not declare.

Figure source and no-synthetic-signal contract. The transient figures (Sections 4.1 and 10) are rendered from the accepted trajectory `output/diagnostics/ieee14_gfm_lock_compare_zeta/adaptive_250s.mat`. Every plotted sample is a raw accepted value. Nothing is smoothed, filtered, decimated, clipped, offset, interpolated or augmented with synthetic noise. Where a display window is applied it is a horizontal range only — no in-window sample is altered or dropped, and an arm whose trajectory ends early is drawn only as far as it ran.

Which result set is presented. The six diagnostic indices of Section 4.1 are published by an *opt-in* post-processing overlay (option `agsi_reference`) that defaults to off, so the previously delivered chronology artefact does not contain them, and one of them cannot be recovered after the fact because the admittance log it needs is built under the same flag. The run above therefore re-executes the present option set with that overlay enabled. The figures and macros in this revision come from the inductive-Thevenin, 17-coordinate result set. It is *not* bit-identical to the earlier ideal-DC or resistive-Thevenin artefacts, because the DC-source model changed deliberately; those revisions are compared by outcome, never by cross-model bit identity. The identity that is established for the present six policy arms is their common pre-disturbance window $[0, 20)$ s: all 42 accepted samples agree exactly before the policies are allowed to diverge. See defect records NUM-2026-08-20-01 and DOC-2026-08-26-03.

Classification. The SG AC/flux equations and the IBR AC/control equations are SOURCE_MAPPED: their structural form is the established one in the converter and machine literature cited where each block is derived [1, 2, 6, 7, 10, 11], and their transcription into this project's state layout is recorded in the project's own equation contract. The fixed 17-state IBR superset, the GFL↔GFM transfer map, the command-delay reduction of Section 6.1 and the DC-source regulator closure are PROJECT_DERIVED. Machine parameters come from the Waterloo IEEE-14 dynamic-data table [4]; the network, branch data and load schedule come from the MATPOWER IEEE 14-bus case [5]; converter parameters, per-unit bases and the disturbance times are otherwise CASE_DEFINED constants of the case file. The predictor and step-subdivision policy are NUMERICAL_METHOD. The four reference-only indices of Section 4.1 are ASSUMED_DIAGNOSTIC and enter no gate. The centre-of-inertia frequency is a project convention, defined explicitly at (56); no external source is claimed for it. These labels are carried from the code provenance fields and are not re-decided here.

2.1 Nomenclature

Table 1 collects the symbols used throughout, grouped by the subsystem that owns them. Symbols are introduced again where they first appear; this table exists so a reader entering at any section can resolve one without searching backwards.

Table 1: Principal symbols.

Symbol	Meaning
System level	
Y_{net}, V	network admittance matrix; stacked complex bus voltages
$I_{\text{bus}}(x, V)$	total device current injection into the network
$g(x, V)$	all-KCL algebraic residual, $Y_{\text{net}}V - I_{\text{bus}}$
x, n_x	composite differential state and its dimension
S_k, E_b	selection operators: $x_k = S_k x$, $V_b = E_b V$
$\mathcal{A}, \mathcal{F}$	active and frozen state index sets, $P_{\mathcal{A}}$ the active projection
m_k	discrete mode label of device k
$h, r(z), J$	step size; coupled Newton residual; its Jacobian
$S_{\text{base}}, f_0, \omega_b$	system power base; nominal frequency; base electrical speed
κ	system-to-machine power base ratio
f_{coi}	inertia-weighted centre-of-inertia frequency, (56)
Ω	damping margin of a configuration's own equilibrium
Switching supervisor	
J_V, J_f	normalised voltage- and frequency-deviation sub-indices
$S(t)$	severity index, $\text{sat}_{[0,1]}(\frac{1}{2}J_V + \frac{1}{2}J_f)$
$\Gamma_{\text{on}}, \Gamma_{\text{off}}$	upper and lower comparator thresholds
$T_{d,\text{on}}, T_{d,\text{off}}$	minimum dwell before promotion and before release
$J_R, J_P, J_{\text{lock}}, J_{\text{SCR}}$	reference-only diagnostic indices; enter no gate
Synchronous generator	
δ, ω	rotor angle; per-unit speed deviation
E'_q, E'_d	transient EMFs (E'_d frozen at the singular limit)
E''_q, E''_d	sub-transient EMFs
T_m, E_{fd}	mechanical torque; field voltage
T_e, I_d, I_q	air-gap torque; stator currents in the machine frame
H, D, R_a	inertia constant; damping; armature resistance
X_d, X'_d, X''_d	direct-axis synchronous, transient and sub-transient reactances
c_d, d_d, c_q, d_q	flux-coupling coefficients, (13)
Inverter-based resource	
i_d, i_q, V_{dc}	filter currents and DC-link voltage (shared power stage)
$\delta_{\text{PLL}}, \xi_{\text{PLL}}$	PLL angle and integrator (GFL only)
ξ_P, ξ_Q	outer active- and reactive-power integrators (GFL)
$\delta_{\text{VSG}}, \omega_{\text{VSG}}$	virtual-machine angle and speed (GFM only)
E	voltage-loop reference state of the Q–V loop (GFM)
ξ_{Vd}, ξ_{Vq}	voltage-loop integrators (GFM)
ξ_{Id}, ξ_{Iq}	inner current-loop integrators (both modes)
v_{cd}, v_{cq}	inner-loop commanded voltages
R_f, L_f, C_{dc}	filter resistance and inductance; DC-link capacitance
E_{dc}, R_{dc}	DC-source EMF and internal resistance (Section 6.3)

Symbol	Meaning
R_{ch}, $V_{dc}^{\max}$	overvoltage-chopper resistance and conduction threshold
I_{dc}	DC-source current, state 17 of the IBR
τ_s, L_s	DC-source circuit time constant L_s/R_{dc} and its inductance
ε, P_r	DC-source regulation at rated converter power, and that power
M, D_v, τ_E, k_Q, k_E	virtual inertia, damping, amplitude time constant and Q–V gains
$I_{\max}$, Π_I	current magnitude limit and the circular limiter
$\mathcal{H}$	conditional anti-windup integrator

3 Disturbance chronology

The scheduled disturbance times are CASE_DEFINED constants of the event structure; the re-close and severity-release instants are outcomes read from the accepted event log. The shunt fault is applied at $t = 85.000$ s and cleared at $t = 85.150$ s (a 0.150 s bus-9 fault, impedance $Z_f = 0.01 + \mathrm{j}0.01$ p.u.); this exact window is taken from the driver and supersedes any wider nominal fault interval.

The six operating periods and the transition instants are drawn in Figure 4, which in this revision is deliberately placed *above the result figures* at the head of Section 10, so that the timeline and the curves can be compared in one view without turning back a page.

4 Automatic GFL/GFM selection: the severity index

Each IBR carries a two-state supervisor that decides whether it should behave as a grid-following current source (GFL) or as a grid-forming voltage source (GFM). The decision is driven by a scalar severity $S(t) \in [0,1]$ built from two normalised sub-indices: a voltage-deviation index J_V and a frequency-deviation index J_f.

Sub-indices. Let $|V_i(t)|$ be the terminal voltage magnitude of IBR i and $V_i^\star$ the healthy per-bus power-flow magnitude at the SG-online operating point (Section 8). Let $f_{\mathrm{coi}}(t)$ be the centre-of-inertia frequency. With bases $\Delta V_{\mathrm{base}} = 0.10$ p.u. and $\Delta f_{\mathrm{base}} = 0.50$ Hz,

$$J_V(t) = \frac{\big|\,|V_i(t)| - V_i^\star\big|}{\Delta V_{\mathrm{base}}}, \qquad J_f(t) = \frac{\big|f_{\mathrm{coi}}(t) - f_0\big|}{\Delta f_{\mathrm{base}}}, \qquad f_0 = 60\ \mathrm{Hz}. \tag{2}$$

Two-term severity (AGSI). The severity is the saturated equal-weight average

$$S(t) = \mathrm{sat}_{[0,1]}\Big(\tfrac{1}{2}J_V(t) + \tfrac{1}{2}J_f(t)\Big), \qquad \mathrm{sat}_{[0,1]}(z) = \min\{1, \max\{0, z\}\}. \tag{3}$$

J_V responds to a local voltage collapse or overshoot; J_f responds to a system-wide frequency excursion such as the one that follows the loss of the SG slack at $t = 20$ s. The equal weighting is a PROJECT_DERIVED choice and is frozen before any result is read.

Hysteresis and dwell. A bare threshold on S would chatter. The supervisor therefore uses an asymmetric two-level comparator with minimum dwell times:

$$\begin{aligned} \text{GFL} \to \text{GFM}: &\quad S \geq \Gamma_{\text{on}} = 0.65 \text{ held for } T_{d,\text{on}} = 0.10\,\text{s}, \\ \text{GFM} \to \text{GFL}: &\quad S \leq \Gamma_{\text{off}} = 0.35 \text{ held for } T_{d,\text{off}} = 1.00\,\text{s}. \end{aligned} \tag{4}$$

The gap $\Gamma_{\text{on}} > \Gamma_{\text{off}}$ gives the comparator its hysteresis band; the asymmetric dwell ($T_{d,\text{on}} \ll T_{d,\text{off}}$) makes the supervisor quick to add grid-forming support and deliberately slow to release it.

Reference-loss guard. A separate guard runs before the dwell logic and takes precedence over it: a breaker opening that would leave an energised island with *no* voltage-forming device at all is **refused**. The network must not spend even one step without a voltage reference, or (1) has nothing to fix the reference angle and the system loses its mathematical gauge condition. The guard is evaluated *before* Newton is called even once, and its effect is measured: an arm that locks all four IBRs in grid-following mode is refused at exactly $t = 20.000\,\text{s}$, the instant of the trip, with `noVoltageFormingSource`. The guard is a feasibility gate, not a promotion rule — it does not itself raise any converter to GFM. In the delivered run the supervisor reaches its maximum of 4 grid-forming units by 4 separate staged commitments, not by a single simultaneous promotion at the trip.

4.1 Decision evidence: every index on one time axis

Figure 1 puts every published index on one page against one shared time axis, so the value of each can be read at the second the supervisor actually commanded a switch. The vertical lines come in *two families* that must not be confused. Dotted lines are the scheduled disturbances (20, 50, 85, 110, 145 s and the achieved reclose); dash–dotted lines are the supervisor's *own* commitments (mode augmentation and release). In this run the two families do not coincide at a single instant, which is exactly why a figure marked only with the disturbance times cannot show when the switching happened.

Panel (a) is the decision variable itself with both thresholds drawn. Panels (b) and (c) are the two terms that form it. Panels (d)–(g) are reference indices that **enter no gate**: they are computed purely after the trajectory is complete, are classified ASSUMED_DIAGNOSTIC, and must not be used to support a readiness claim. Two properties are worth stating so the figure is not misread: J_f and J_R are *system* quantities built from the centre-of-inertia frequency, so they are single traces rather than four; and J_{lock} is exactly zero on a grid-forming branch by construction, because a GFM branch has no PLL to lose lock — not because the quantity could not be measured. No aggregate index is formed from these four, by design: an aggregate would sit one step away from being a decision variable.

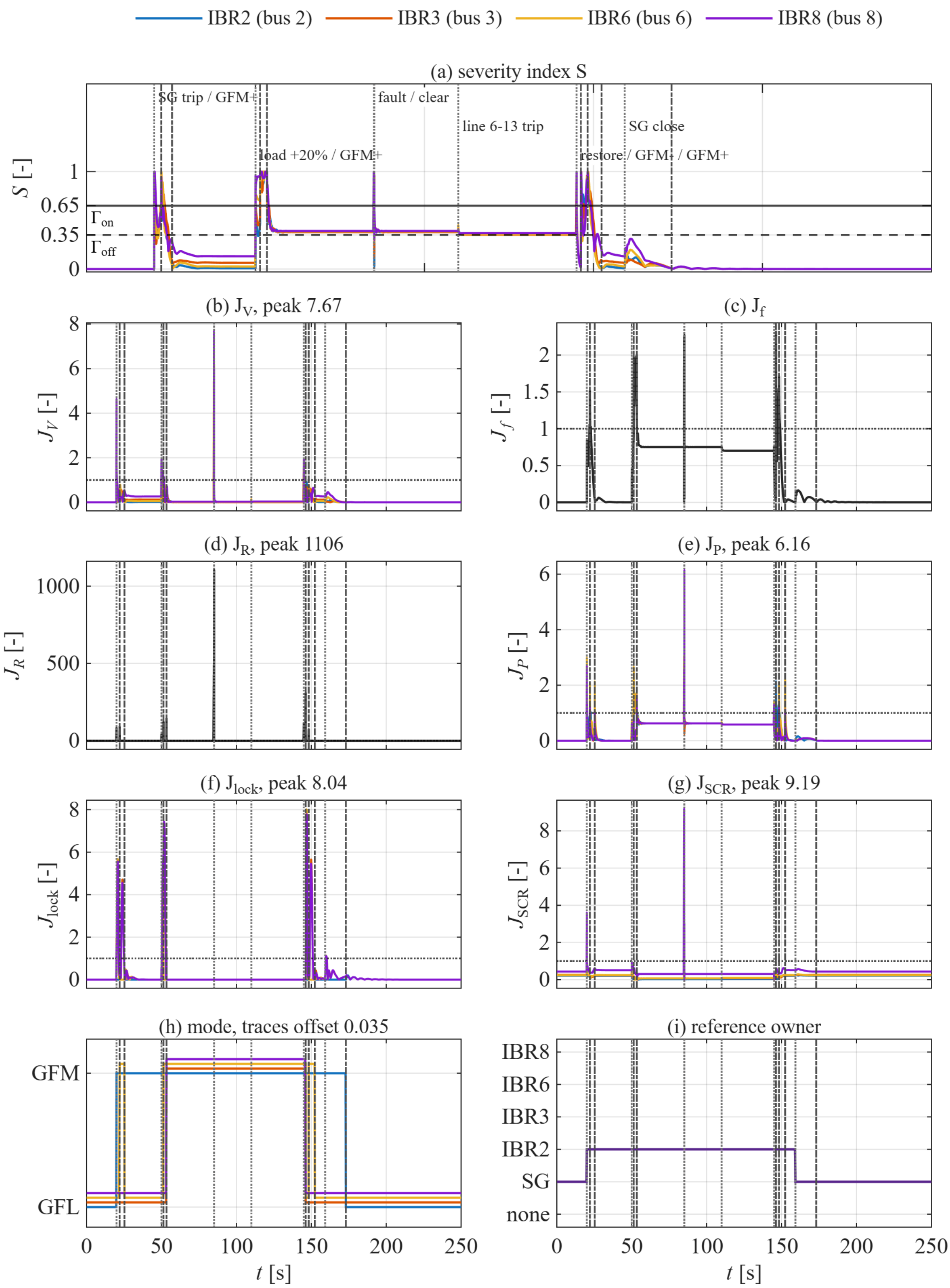

Figure 1: Switching-decision evidence; every panel shares one time axis and one set of event lines. (a) Severity $S = \mathrm{sat}_{[0,1]}(0.5J_V + 0.5J_f)$ with the thresholds $\Gamma_{on} = 0.65$ and $\Gamma_{off} = 0.35$. (b) J_V. (c) J_f (system). (d) J_R (system). (e) J_P. (f) J_{lock}. (g) J_{SCR}. (h) per-device GFL/GFM mode. (i) reference-angle owner. Dotted lines: scheduled disturbances. Dash–dotted lines: supervisor commitments. Panels (a)–(c) decide; panels (d)–(g) are reference-only and enter no gate.

The two windows in which the switching actually happens are enlarged in Figures 2 and 3.

In those two figures both line families are fully labelled, so the index values at the instant of each command can be read off directly.

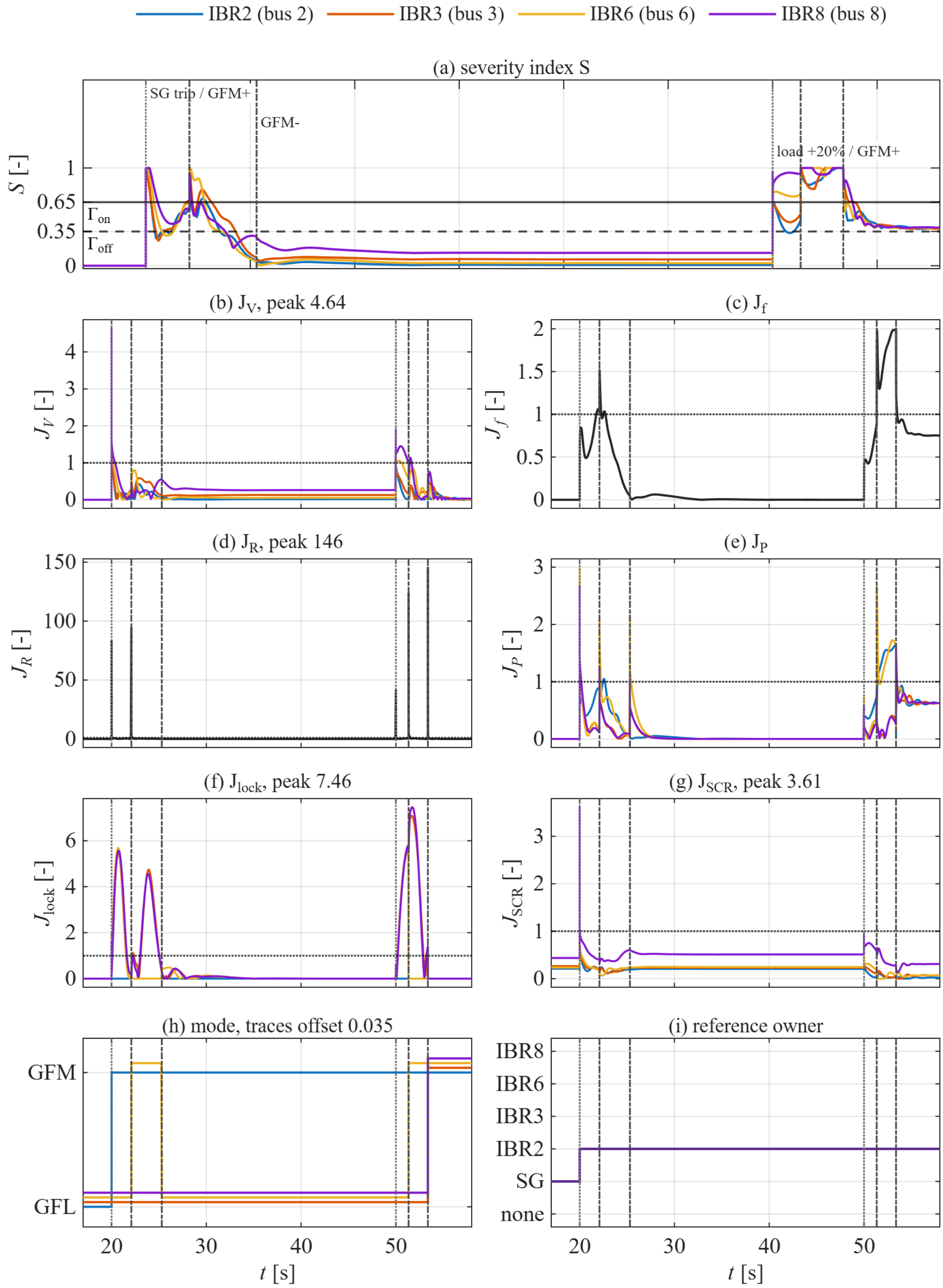

Figure 2: Islanding window $t \in [17, 58]$ s, covering the SG trip, the first two mode augmentations, one release and the load step. Panels and line families are identical to Figure 1; only the time range differs. No sample is dropped or altered.

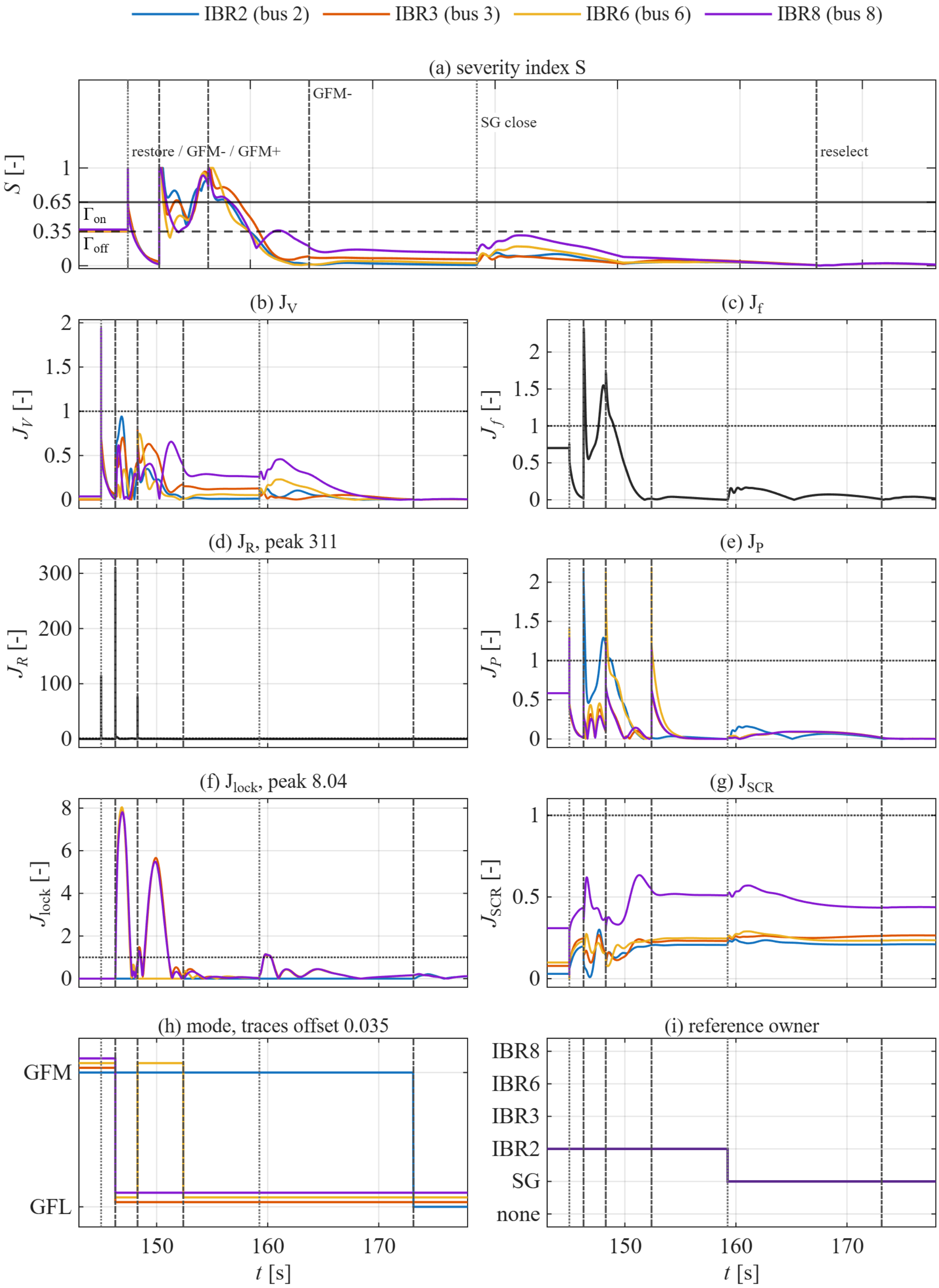

Figure 3: Reclose and hand-back window $t \in [143, 178]$ s, showing the mode releases, one compensating augmentation, the reclose at $t = 159.252$ s, and the return of reference-angle ownership to the SG. Panels and line families are identical to Figure 1.

5 Synchronous generator: sixth-order model and its proof

The SG is a sixth-order round-rotor machine of the structure IEEE Std 1110 recommends for stability studies of this kind [1], with state

$$x_{\mathrm{sg}} = [\,\delta,\ \omega,\ E'_q,\ E'_d,\ E''_q,\ E''_d\,]^\top, \qquad u_{\mathrm{sg}} = [\,T_m,\ E_{fd}\,]^\top, \tag{5}$$

where δ is the rotor angle, ω the per-unit speed deviation (slip), E'_q, E'_d the transient EMFs, and E''_q, E''_d the subtransient EMFs. The two control inputs T_m (mechanical torque) and E_{fd} (field voltage) are held at the constant values determined by the mixed-network equilibrium solve; they are not re-solved each step.

Attribution of the machine structure. Two lineages are easy to conflate here and are kept apart deliberately. The *structural class* — a round-rotor machine carrying transient and sub-transient flux pairs on both axes — is the one IEEE Std 1110 recommends, and that is what the reference above supports [1]. The *particular algebraic form* in which such a machine is coded in industry tools descends from a separate line of work on machine-model equivalence and is not an IEEE 1110 object; the two are not interchangeable and are not treated as such here. IEEE Std 1110 also does not itself supply the swing equation: for that it defers to the standard textbook treatments [2, 3], which is the form used in (6)–(7). No page number is quoted from the standard, because two byte-distinct copies of it circulate and their pagination differs; a page citation that cannot be reproduced is worse than none.

Where the machine numbers come from. The reactances, time constants and inertia of the machine at the reference bus are CASE_DEFINED values taken from the Waterloo technical report's IEEE-14 dynamic-data table [4], mapped onto the flux parameterisation of (13), with $H = 2.5\,\mathrm{s}$ and $D = 1.0\,\mathrm{p.u.}$ on the 100 MVA system base. The network topology, branch data and load schedule are the MATPOWER distribution of the IEEE 14-bus case, itself converted from the IEEE common data format [5]. Neither source is re-derived here and no value from either is adjusted; Section 8 reports the power flow they produce.

5.1 Order-by-order derivation

The connected-machine equations are, term for term as coded,

$$\dot{\delta} = \omega_0\,\omega, \tag{6}$$

$$2H\,\dot{\omega} = T_m - T_e - D\,\omega, \tag{7}$$

$$T'_{d0}\,\dot{E}'_q = E_{fd} + c_d E''_q - d_d E'_q, \tag{8}$$

$$T'_{q0}\,\dot{E}'_d = c_q E''_d - d_q E'_d, \tag{9}$$

$$T''_{d0}\,\dot{E}''_q = E'_q - E''_q - (X'_d - X''_d)\,I_d, \tag{10}$$

$$T''_{q0}\,\dot{E}''_d = E'_d - E''_d + (X'_q - X''_q)\,I_q, \tag{11}$$

with $\omega_0 = 2\pi f_0$. The air-gap electrical torque that closes the swing equation (7) is

$$T_e = V_d I_d + V_q I_q + R_a\,(I_d^2 + I_q^2), \tag{12}$$

and the four flux-coupling coefficients are defined from the reactance ladder as

$$c_d = \frac{X_d - X'_d}{X'_d - X''_d}, \quad d_d = \frac{X_d - X''_d}{X'_d - X''_d}, \quad c_q = \frac{X_q - X'_q}{X'_q - X''_q}, \quad d_q = \frac{X_q - X''_q}{X'_q - X''_q}. \tag{13}$$

What each order is. Equations (6)–(7) are the mechanical (2nd-order) swing: angle and speed. Equations (8)–(9) are the transient (rotor field / q-axis damper) fluxes on the slow time constants T'_{d0}, T'_{q0}. Equations (10)–(11) are the subtransient (damper) fluxes on the fast time constants T''_{d0}, T''_{q0}. The armature-reaction terms $-(X'_d - X''_d)I_d$ and $+(X'_q - X''_q)I_q$ couple the fast fluxes to the stator current.

Proof anchor — the $d - c = 1$ identity. From (13),

$$d_d - c_d = \frac{(X_d - X''_d) - (X_d - X'_d)}{X'_d - X''_d} = \frac{X'_d - X''_d}{X'_d - X''_d} = 1, \qquad \text{and identically } d_q - c_q = 1. \quad (14)$$

Consider the open-circuit steady state ($I_d = I_q = 0$, $\dot{x} = 0$). Then (10) gives $E''_q = E'_q$, and substituting into (8),

$$0 = E_{fd} + c_d E'_q - d_d E'_q = E_{fd} - (d_d - c_d)E'_q = E_{fd} - E'_q \implies E'_q = E_{fd}. \quad (15)$$

So the identity (14) is exactly what makes the open-circuit field EMF equal the applied field voltage — the standard consistency check for this flux parameterisation, and the reason the coefficients are written as ratios (13) rather than as free constants. The same argument on the q-axis gives $E'_d = 0$ at open circuit when E_{fd} has no q-axis analogue.

5.2 Algebraic stator interface

The stator currents are *not* reverse-derived from power; they are solved from the subtransient EMFs behind subtransient reactance. Projecting the complex terminal voltage V onto the machine dq frame (V_d, V_q) and writing $r_d = V_d - E''_d$, $r_q = V_q - E''_q$,

$$\det = X''_d X''_q + R_a^2, \qquad I_d = \frac{-R_a r_d - X''_q r_q}{\det}, \qquad I_q = \frac{X''_d r_d - R_a r_q}{\det}, \quad (16)$$

which is the exact inverse of the subtransient stator relations $V_d = E''_d - R_a I_d + X''_q I_q$, $V_q = E''_q - R_a I_q - X''_d I_d$. The electrical torque (12) uses these same I_d, I_q.

5.3 The singular limit that freezes E'_d

For the machine data used here the q-axis transient open-circuit time constant is set to its singular limit $T'_{q0} = 0$ (audited round-rotor case). Then (9) is no longer differential: it degenerates to the algebraic constraint

$$0 = c_q E''_d - d_q E'_d. \quad (17)$$

For this machine $c_q = 0,\ d_q = 1$, so $E'_d \equiv 0$. The code detects $T'_{q0} = 0$ exactly and *removes* E'_d (state index 4) from the integrated set, holding it at its algebraic value 0. The SG therefore contributes **five** dynamic states $[\delta, \omega, E'_q, E''_q, E''_d]$, not six. This is a SOURCE_DEFINED singular limit, not a numerical convenience: it is the $T'_{q0} \to 0$ reduction of the same equations.

5.4 Breaker-open coast

When the SG breaker opens at $t = 20$ s the machine does not vanish from the state vector. Its network injection becomes zero ($I_d = I_q = 0$, $T_e = 0$) and its fluxes decay on the open-circuit branch of (8)–(11) (the armature-reaction terms drop out because $I_d = I_q = 0$); the swing (6)–(7) coasts on a frozen mechanical torque T_m. The map that reports how many states the SG integrates is context-independent: it returns $[\,1, 2, 3, 5, 6\,]$ whether the breaker is closed or open. Consequently the loss of the SG slack does *not* change the SG's own state count; the composite dimension changes only because the IBRs switch mode (Section 6).

6 Inverter-based resource: the dual-mode 17-state model

Each IBR is a single device that can present either of two controllers on a common power stage. The full state is a fixed 17-vector,

$$x_{\mathrm{IBR}} = \big[\underbrace{i_d,\, i_q,\, V_{dc}}_{\substack{(3)\\ \text{always live}}} \;\big|\; \underbrace{x_{\mathrm{GFL}}\ (6)}_{\substack{\text{indices 4–9}\\ \text{live only in GFL}}} \;\big|\; \underbrace{x_{\mathrm{GFM}}\ (7)}_{\substack{\text{indices 10–16}\\ \text{live only in GFM}}} \;\big|\; \underbrace{I_{dc}}_{\substack{\text{index 17}\\ \text{always live}}}\ \big]^{\top} \in \mathbb{R}^{17}. \tag{18}$$

The plant block is $x_{\mathrm{pl}} = (i_d, i_q, V_{dc}, I_{dc})$ and it is not contiguous: the DC-source current sits at index 17, after the two controller blocks, because it was added to a layout whose indices 1–16 were already published. Appending rather than inserting keeps every one of those indices meaning exactly what it meant before, which is why the ranges 4–9 and 10–16 in this section are unchanged from earlier revisions. A test asserts that all sixteen earlier names are still in their original positions.

The two controllers never run at the same time. Equation (18) is a fixed *superset* of coordinates, not a list of simultaneously live states, and this is the single most important thing to read correctly about the device. A converter is *either* grid-following *or* grid-forming at any instant, never both and never partly each: one controller block is integrated while the other is held frozen at the value it carried when it last surrendered the duty, and the two can exchange the duty only at a switching instant, through the transfer of Section 6.6. There is no blend, no weighted average of the two control laws, and no interval on which both sets of integrators advance. The consequence for any later reading of the state vector is concrete: an index in the frozen block is not a measurement of anything during that interval, so a quantity built from a fixed index across a mode change is not a physical signal.

The shorthand $x_{\mathrm{IBR}} = [x_{\mathrm{GFM}}\ x_{\mathrm{GFL}}]$ is the same object with the shared plant block x_{pl} made explicit and the two controller blocks kept in their code order. The controller blocks are

$$x_{\mathrm{GFL}} = [\,\delta_{\mathrm{PLL}},\, \xi_{\mathrm{PLL}},\, \xi_P,\, \xi_Q,\, \xi_{Id},\, \xi_{Iq}\,], \tag{19}$$

$$x_{\mathrm{GFM}} = [\,\delta_{\mathrm{VSG}},\, \omega_{\mathrm{VSG}},\, E,\, \xi_{Vd},\, \xi_{Vq},\, \xi_{Id},\, \xi_{Iq}\,]. \tag{20}$$

Only a projection of (18) is integrated at any instant. The active-index map returns

$$\mathcal{A}_{\mathrm{GFL}} = \{1{:}9,\ 17\}, \qquad \mathcal{A}_{\mathrm{GFM}} = \{1{:}3,\ 10{:}16,\ 17\}, \qquad \mathcal{A}_{\mathrm{tripped}} = \varnothing, \tag{21}$$

i.e. 10 active states in GFL, 11 in GFM, none when tripped. The intersection $\mathcal{A}_{\mathrm{GFL}} \cap \mathcal{A}_{\mathrm{GFM}} = \{1, 2, 3, 17\}$ is exactly the shared plant, and the union is all of $\{1{:}17\}$: the plant is always live, and every controller index belongs to one branch alone. The active count is therefore 10 or 11, never 17 — the superset dimension is never the dynamic dimension. With one SG contributing six stored states, the composite state vector of this study therefore has $6 + 4 \times 17 = 74$ rows, which is exactly the row count of the stored trajectory.

6.1 Reduced command-delay states

The source block diagram carries one first-order actuation lag per axis, $T_d\,\dot{V}_{\mathrm{del}} = v_{\mathrm{cmd}} - V_{\mathrm{del}}$, whose physical time constant is the digital-converter delay $T_d = 1.5/f_{\mathrm{sw}} \approx 0.3$ ms at $f_{\mathrm{sw}} = 5$ kHz. That is more than two orders of magnitude below the phasor step used here, so by singular perturbation the fast lag collapses onto its slow manifold $V_{\mathrm{del}} = v_{\mathrm{cmd}}$: the two delay states per branch are removed algebraically and the filter-current equations use the commanded voltage v_{cd}, v_{cq} directly. This is a standard model-order reduction that leaves both the equilibrium and the retained-state dynamics unchanged, and it removes a pole near 530 Hz that an RMS phasor network model cannot represent in the first place. The reduction takes the dual-mode superset from 20 to 16 states and each standalone branch from 12 to 10; the DC-source current of Section 6.3 then adds one, giving the 17 of (18) and 11 per standalone branch. It is PROJECT_DERIVED and owner-set; see defect record TD-2026-08-12-01. Report revisions before this one printed the pre-reduction 20-state layout, which the executed model no longer has.

Two operators recur below; their structural role in a limited converter is the standard one [8], while the particular directional anti-windup realisation is PROJECT_DERIVED. $\Pi_I(a, b; I_{\max})$ is the *circular* current limiter: if $r = \sqrt{a^2 + b^2} > I_{\max}$ it scales $(a, b) \mapsto (a, b)\, I_{\max}/r$, else it is the identity; its rejected part is the limiter residual $r_i = (a, b) - \Pi_I(a, b)$. $\mathcal{H}(e, \gamma, r_i, \mathrm{sat})$ is the conditional anti-windup integrator: it returns 0 when the reference is clipped and the error would wind further (sat and $\gamma\, e\, r_i > 0$), and e otherwise. The instantaneous injection is $I = (i_d + \mathrm{j} i_q)e^{\mathrm{j}\theta}/\kappa$, $S = V\bar{I}$, $P_{\mathrm{inv}} = \kappa\Re S$, $Q_{\mathrm{inv}} = \kappa\Im S$, with $\kappa = S_{\mathrm{base}}/M_{\mathrm{base}}$ and $(v_d, v_q) = \Re, \Im\,(Ve^{-\mathrm{j}\theta})$.

6.2 Shared power stage (states 1–3)

The L-filter current dynamics and DC-link are common to both modes and are always active. Written one state per line in index order:

$$\frac{L}{\omega_b}\dot{i}_d = v_{cd} - v_d - R\, i_d + \omega L\, i_q, \qquad d\text{-axis filter current} \qquad (22)$$

$$\frac{L}{\omega_b}\dot{i}_q = v_{cq} - v_q - R\, i_q - \omega L\, i_d, \qquad q\text{-axis filter current} \qquad (23)$$

$$\dot{V}_{dc} = \frac{1}{C}\left[\frac{E_{dc}-V_{dc}}{R_{dc}} - \frac{P_{ac}}{V_{dc}} - \frac{\max(0,\, V_{dc} - V_{dc}^{\max})}{R_{ch}}\right], \qquad \text{DC-link voltage} \qquad (24)$$

with $R = R_f$, $L = L_f$, $C = C_{dc}$, $\omega_b = 2\pi f_b$, and v_{cd}, v_{cq} the inner-loop commanded voltages defined per branch below. The DC-link row is derived in Section 6.3; $P_{ac} = v_{cd}i_d + v_{cq}i_q$ there is the converter-side power, that is the bus power plus the filter loss, which is what the DC bus actually has to supply. The commanded voltage appears here rather than a delayed command because of the reduction of Section 6.1. *Proof of each row.* Rows (1)–(2) are Kirchhoff's voltage law across the series R_f, L_f filter in the rotating frame; the cross terms $\pm\omega L$ are the frame-rotation EMF, and the rotation rate ω is supplied by whichever controller is active (the PLL in GFL, the VSG swing in GFM) — this is the only place the plant equations feel the mode. Row (3) is the PROJECT_DERIVED DC-source closure derived in Section 6.3: the source publishes the energy balance $C\dot{V}_{dc} = I_{dc} - P_{ac}/V_{dc}$ but not the law for I_{dc}, and the coded current is the Thévenin law (25), $I_{dc} = (E_{dc} - V_{dc})/R_{dc}$, less the chopper draw (30). Nothing in row (3) cancels: P_{ac} drives the coordinate and C_{dc} scales it, so both enter the residual and the Jacobian. Setting $\dot{V}_{dc} = 0$ with the chopper off gives $V_{dc}^2 - E_{dc}V_{dc} + R_{dc}P_{ac} = 0$, whose upper root is $V_{dc} = V_{dc}^0$ at the dispatched loading $P_{ac} = P_{ac}^0$ — that is what fixes E_{dc} in (26) — and moves with P_{ac} at every other loading. The measured excursions are reported at the end of Section 6.3.

6.3 Non-ideal DC source

The source publishes the DC-link energy balance $C\dot{V}_{dc} = I_{dc} - P_{ac}/V_{dc}$ but not the law for I_{dc}, so the law is PROJECT_DERIVED and has to be stated and justified rather than assumed. Any

DC-side source — a rectifier, a battery, a photovoltaic string in its voltage-source region, or a DC feeder — is to first order an EMF behind its internal plus cable resistance, which gives

$$\tau_s \dot{I}_{dc} = \frac{E_{dc} - V_{dc}}{R_{dc}} - I_{dc}, \qquad \tau_s = \frac{L_s}{R_{dc}}, \tag{25}$$

in which I_{dc} is a *state*: that circuit has inductance, so the current it actually delivers chases its own static characteristic rather than equalling it instantaneously. Setting $\tau_s = 0$ recovers the algebraic law $I_{dc} = (E_{dc} - V_{dc})/R_{dc}$, and that limit is exact rather than approximate — see the degeneracy statement below — so a reader comparing this section with an earlier revision is comparing one model with its own limit, not two models. Equation (25), the energy balance above it and the overvoltage chopper of (30) below are together the DC-link rows of the shared power stage.

E_{dc} **is forced, not chosen.** Requiring the dispatched point to be an equilibrium, $\dot{V}_{dc} = 0$ at $V_{dc} = V_{dc}^0$ and $P_{ac} = P_{ac}^0$, leaves exactly one admissible value,

$$E_{dc} = V_{dc}^0 + \frac{R_{dc} P_{ac}^0}{V_{dc}^0}, \qquad P_{ac}^0 = \kappa P_{\text{ref}} + R_f \left(i_{d0}^2 + i_{q0}^2\right), \tag{26}$$

in closed form, because at the initial point both branches set $i_{d0} = \kappa P_{\text{ref}}/|V_0|$ and $i_{q0} = -\kappa Q_{\text{ref}}/|V_0|$ with $\theta = \angle V_0$, so $v_d = |V_0|$, $v_q = 0$, and the two ωL cross terms of $v_{cd} i_d + v_{cq} i_q$ cancel identically. Two consequences matter. First, the sourced AC initial condition and therefore the power flow of Section 8 are untouched: the measured $\dot{V}_{dc}(0)$ is zero to round-off. Second, because (26) is evaluated from the same $P_{\text{ref}}, Q_{\text{ref}}, V_0, R_f$ in both branches, E_{dc} is identical in GFL and GFM, so a run-time transfer introduces no step in $\dot{V}_{dc}$ and not merely none in V_{dc}.

R_{dc} **from a declared regulation, bounded by a reachability requirement.** Let ε be the DC-source regulation, the fractional DC voltage rise from the dispatched point to no load at rated converter power P_r, so that $R_{dc} = \varepsilon (V_{dc}^0)^2/P_r$. The equilibrium of (25) against a constant-power draw P satisfies $V_{dc}^2 - E_{dc} V_{dc} + R_{dc} P = 0$, which has a real solution only while

$$E_{dc}^2 \;\geq\; 4 R_{dc} P_{\max}, \qquad P_{\max} = V_{\max} I_{\max}, \tag{27}$$

so requiring margin m in (27) bounds ε from above. At $m = 2$ with $V_{\max} = 1.06\,\text{p.u.}$, $I_{\max} = 1.2\,\text{p.u.}$ and the dispatched P_{ref} of this study the bound is $\varepsilon \lesssim 0.115$. The declared value is $\varepsilon = 0.10$, which sits inside that bound with realised margin 2.07–2.15 across the four links at the dispatch whose modal table Section 9 prints, and coincides with the usual ten per cent regulation of a DC link fed through a boost stage or a feeder. It is *not* chosen to suit the integrator step; see the stiffness statement below.

τ_s **comes from a filter criterion, with a bound on each side.** Linearising the two DC rows about the dispatched point, chopper off, gives the 2×2 block

$$A_{dc} = \begin{bmatrix} \dfrac{P_{ac}^0}{C (V_{dc}^0)^2} & \dfrac{1}{C} \\ -\dfrac{1}{\tau_s R_{dc}} & -\dfrac{1}{\tau_s} \end{bmatrix}, \quad \det A_{dc} = \frac{K}{\tau_s}, \;\; \operatorname{tr} A_{dc} = a - \frac{1}{\tau_s}, \tag{28}$$

with $K = \frac{1}{C}\left(\frac{1}{R_{dc}} - \frac{P_{ac}^0}{(V_{dc}^0)^2}\right)$ and $a = \frac{P_{ac}^0}{C(V_{dc}^0)^2}$. Two conditions bound τ_s without reference to any reported quantity. Positive determinant requires $R_{dc} < (V_{dc}^0)^2/P_{ac}^0$, which is the same constant-power-load condition the algebraic source had; adding the state does not relax it. Negative trace requires

$$\tau_s < \frac{C(V_{dc}^0)^2}{P_{ac}^0}, \tag{29}$$

which is *new*: a source that reacts too slowly cannot hold a constant-power load. Between those bounds τ_s is fixed by the maximally-flat criterion $\zeta = 1/\sqrt{2}$ on the L–C pair, the ordinary way a filter of this shape is specified, which with $x = 1/\tau_s$ and $\zeta = (x-a)/(2\sqrt{Kx})$ has the closed form $\tau_s = 1/y^2$, $y = \zeta^\star\sqrt{K} + \sqrt{\zeta^{\star 2}K + a}$. On this device that returns $\tau_s = 5.00$ ms, the realised damping is 0.707 at $P_{ac}^0 = 0.45$ p.u. and 0.708 at 0.90 p.u. — so one declared component value serves the whole dispatch range — and the pair sits at -97.7 ± 97.7 s^{-1}, about 15.5 Hz. **Degeneracy:** as $\tau_s \to 0$ the pair separates into a fast root near $-1/\tau_s$ and a slow root at $-K$, which is exactly the eigenvalue of the algebraic-source model; this is verified numerically and falls linearly with τ_s.

What replaces the run-away bound. With an algebraic source the link obeyed $V_{dc} \leq \max(V_{dc}(0), E_{dc})$ whenever $P_{ac} \geq 0$, by a sign argument on a single row. That argument does not survive the extra state, because an inductive source overshoots: at $\zeta = 1/\sqrt{2}$ the step response passes its target once by about four per cent before settling. What does survive is the mechanism rather than the inequality — the source row of (25) drives I_{dc} *down* whenever V_{dc} exceeds E_{dc}, and (28) is stable, so the pair returns. At this dispatch E_{dc} ranges 1.0269–1.0448 p.u. over the four links and still lies below the chopper threshold $V_{dc}^{\max} = 1.10$ p.u., but that is now a statement about the settling point and not a bound on the excursion. The overvoltage chopper is a braking resistor conducting in its linear region,

$$I_{ch} = \frac{\max\left(0,\, V_{dc} - V_{dc}^{\max}\right)}{R_{ch}}, \tag{30}$$

continuous and Lipschitz, non-differentiable only on $V_{dc} = V_{dc}^{\max}$ in exactly the way Π_I is, so it needs no separate smoothness argument and is covered by Remark 2.

Why the chopper is not ornamental. Everything above assumes the converter *exports*. A converter driven into *absorption* — held at its current limit while the island voltage and angle run away from it — turns the load term of the link equation into a charging term $+|P_{ac}|/V_{dc}$. Its equilibrium is then the upper root of $V_{dc}^2 - E_{dc}V_{dc} + R_{dc}P_{ac} = 0$ with $P_{ac} < 0$, which lies above E_{dc} and, for the powers a saturated current limiter permits, above $V_{dc}^{\max}$. The chopper is what arrests it. That is not a hypothetical condition: it arises among the five control-policy arms of this study whenever the island runs away from a converter faster than the converter can follow. Whether it conducted on a given trajectory is decided afterwards from $\max_t V_{dc}$ against $V_{dc}^{\max}$, an exact test because the conduction condition depends on V_{dc} alone; the per-arm outcome is reported with the trajectories rather than asserted here, and the chopper is never presented as protection that was proved unnecessary.

The stiffness this exposes is reported, not designed around. The slow root of (28), which is also the whole spectrum of the algebraic-source limit, is

$$\lambda_{dc} = -\frac{1}{C}\left(\frac{1}{R_{dc}} - \frac{P_{ac}^0}{(V_{dc}^0)^2}\right), \tag{31}$$

whose second term is the negative incremental resistance of a constant-power load, so stability requires $R_{dc} < (V_{dc}^0)^2/P_{ac}^0$ — a condition this configuration satisfies by a wide factor. With $C = C_{dc} = 0.10$ and $R_{dc} = 0.10$ the mode is near $-95\ \mathrm{s}^{-1}$, a time constant of about 10 ms, so $|\lambda_{dc}|\,\Delta t \approx 4.8$ at the nominal phasor step $\Delta t = 0.05$ s. The implicit trapezoidal rule is A-stable there, but the mode is under-resolved at the nominal step and the adaptive controller has to reduce Δt where it is excited. ε was deliberately not enlarged to bring $|\lambda_{dc}|\Delta t$ near unity: choosing a physical parameter to suit a step size is reverse-engineering it from numerical convenience. The honest consequence is the reverse of a convenience — a *firmer* DC source, ε of order 0.01 for a battery at the terminals, is stiffer still and belongs to an EMT study rather than to a 50 ms phasor step. That is a limit of this class of simulation, stated here rather than hidden by an idealisation.

What this replaced, and why. Revisions of this report before the present one carried a closure with an exact instantaneous power feed-forward, $I_{dc} = P_{ac}/V_{dc} + (C/T_{dc})(V_{dc}^0 - V_{dc})$, which cancels the AC loading identically and collapses the link to $\dot{V}_{dc} = (V_{dc}^0 - V_{dc})/T_{dc}$, with T_{dc} a declared restoration time constant no longer used on this path. Two measured consequences identify it as an ideal DC source. Its own initial condition $V_{dc}(0) = V_{dc}^0$ is the equilibrium of that scalar equation, so V_{dc} stayed at 1.000000000000 p.u. for every accepted sample of every converter over the whole 250 s run, range exactly zero — the coordinate carried no information. And because C cancels, the declared capacitance C_{dc} never entered any residual. Both are removed by (25)–(30), in which C_{dc} appears for the first time in (31) and V_{dc} is a live coordinate driven by P_{ac}.

6.4 GFL (states 4–9)

The auxiliary algebraic quantities are the power errors $e_P = \kappa u_1 - P_{\mathrm{inv}}$, $e_Q = \kappa u_2 - Q_{\mathrm{inv}}$, the raw and clipped current references

$$i_{d,\mathrm{ref}}^{\mathrm{raw}} = k_{pP}e_P + k_{iP}\xi_P, \quad i_{q,\mathrm{ref}}^{\mathrm{raw}} = -(k_{pQ}e_Q + k_{iQ}\xi_Q), \quad [i_{d,\mathrm{ref}}, i_{q,\mathrm{ref}}] = \Pi_I(\cdot; I_{\max}), \tag{32}$$

and the inner voltage commands $v_{cd} = v_d + Ri_d - \omega L i_q + k_{pI}(i_{d,\mathrm{ref}} - i_d) + k_{iI}\xi_{Id}$, $v_{cq} = v_q + Ri_q + \omega L i_d + k_{pI}(i_{q,\mathrm{ref}} - i_q) + k_{iI}\xi_{Iq}$. The six GFL states then evolve, one per line in index order:

$$\dot{\delta}_{\mathrm{PLL}} = k_{p,\mathrm{PLL}}v_q + k_{i,\mathrm{PLL}}\xi_{\mathrm{PLL}}, \qquad \text{PLL angle;} \quad \omega = 1 + \dot{\delta}_{\mathrm{PLL}}/\omega_b \tag{33}$$

$$\dot{\xi}_{\mathrm{PLL}} = v_q, \qquad \text{PLL integrator} \tag{34}$$

$$\dot{\xi}_P = \mathcal{H}(e_P,\ k_{iP},\ r_{i,d},\ \mathrm{sat}), \qquad \text{active-power integrator} \tag{35}$$

$$\dot{\xi}_Q = \mathcal{H}(e_Q,\ -k_{iQ},\ r_{i,q},\ \mathrm{sat}), \qquad \text{reactive-power integrator} \tag{36}$$

$$\dot{\xi}_{Id} = i_{d,\mathrm{ref}} - i_d, \qquad d\text{-current integrator} \tag{37}$$

$$\dot{\xi}_{Iq} = i_{q,\mathrm{ref}} - i_q, \qquad q\text{-current integrator} \tag{38}$$

Proof of each state. (4)–(5) are the synchronous-reference-frame PLL: it drives the measured q-axis voltage v_q to zero, so at lock $\dot{\xi}_{\mathrm{PLL}} = v_q = 0$ and the frame is aligned with the terminal voltage; the loop frequency is $\omega = 1 + \dot{\delta}_{\mathrm{PLL}}/\omega_b$. (6)–(7) are the outer P/Q PIs: at equilibrium $\dot{\xi}_P = e_P = 0$ forces $P_{\mathrm{inv}} = \kappa u_1$ and $\dot{\xi}_Q = e_Q = 0$ forces $Q_{\mathrm{inv}} = \kappa u_2$, i.e. the IBR meets its P, Q set-points; the $\mathcal{H}$ wrapper freezes them only while the current reference is on the $I_{\max}$ circle. (8)–(9) are the inner current PIs: $\dot{\xi}_{Id} = i_{d,\mathrm{ref}} - i_d = 0$ makes the filter current track its reference (and symmetrically on q). The two command-lag states that the source diagram places between v_{cd}, v_{cq} and the filter are not integrated here; they are on their slow manifold by Section 6.1, so $V_{d,\mathrm{del}} = v_{cd}$ and $V_{q,\mathrm{del}} = v_{cq}$ identically — which is the same steady-state relation the lag would have given, reached without carrying an unresolvable fast pole. The GFL branch is a *current source*: it needs the PLL to know the grid angle, so it cannot by itself define a voltage on a dead network.

6.5 GFM (states 10–16): voltage source, VSG

The grid-forming branch replaces the PLL by a virtual-synchronous-generator swing and a Q–V voltage-forming loop. The voltage references are $v_{d,\mathrm{ref}} = E$, $v_{q,\mathrm{ref}} = 0$; the voltage errors are $e_{vd} = E - v_d$, $e_{vq} = -v_q$; the raw/clipped current references are $i^{\mathrm{raw}}_{d,\mathrm{ref}} = k_{pV} e_{vd} + k_{iV}\xi_{Vd}$, $i^{\mathrm{raw}}_{q,\mathrm{ref}} = k_{pV} e_{vq} + k_{iV}\xi_{Vq}$, $[i_{d,\mathrm{ref}}, i_{q,\mathrm{ref}}] = \Pi_I(\cdot; I_{\max})$; and the inner commands v_{cd}, v_{cq} are as in the GFL branch but with $\omega \to \omega_{\mathrm{VSG}}$. The seven GFM states evolve, one per line in index order:

$$
\begin{aligned}
\dot{\delta}_{\mathrm{VSG}} &= \omega_b(\omega_{\mathrm{VSG}} - 1), && \text{VSG angle} && (39)\\
M\,\dot{\omega}_{\mathrm{VSG}} &= \kappa u_1 - P_{\mathrm{inv}} - D_v(\omega_{\mathrm{VSG}} - 1), && \text{VSG swing} && (40)\\
\tau_E\,\dot{E} &= k_Q(\kappa u_2 - Q_{\mathrm{inv}}) - k_E(E - E_{\mathrm{ref}}), && Q\text{–}V \text{ voltage state} && (41)\\
\dot{\xi}_{Vd} &= \mathcal{H}(e_{vd},\, k_{iV},\, r_{i,d},\, \mathrm{sat}), && d\text{-voltage integrator} && (42)\\
\dot{\xi}_{Vq} &= \mathcal{H}(e_{vq},\, k_{iV},\, r_{i,q},\, \mathrm{sat}), && q\text{-voltage integrator} && (43)\\
\dot{\xi}_{Id} &= i_{d,\mathrm{ref}} - i_d, && d\text{-current integrator} && (44)\\
\dot{\xi}_{Iq} &= i_{q,\mathrm{ref}} - i_q, && q\text{-current integrator} && (45)
\end{aligned}
$$

Proof of each state. (10)–(11) are the VSG: the swing gives the converter its *own* angle from active-power balance, $M\dot{\omega}_{\mathrm{VSG}} = \kappa u_1 - P_{\mathrm{inv}} - D_v(\omega_{\mathrm{VSG}} - 1)$, so at equilibrium $\omega_{\mathrm{VSG}} = 1$ and $P_{\mathrm{inv}} = \kappa u_1$; because the angle is generated internally there is *no PLL* and the branch can energise a passive network. (12) is the Q–V loop whose state E *is* the voltage-loop reference $v_{d,\mathrm{ref}} = E$: at equilibrium $E = E_{\mathrm{ref}} + (k_Q/k_E)(\kappa u_2 - Q_{\mathrm{inv}})$, a droop between reactive set-point and formed voltage. (13)–(14) are the voltage PIs that drive $v_d \to E$, $v_q \to 0$ (so the terminal voltage aligns with the VSG angle at magnitude E); (15)–(16) the inner current PIs. As in the GFL branch the two command lags are on their slow manifold and are not integrated (Section 6.1). The anti-windup $\mathcal{H}$ again holds the voltage integrators only while the current reference is saturated. The GFM branch is a *voltage source*: this is why at least one grid-forming converter must exist for the island to have a reference angle at all.

6.6 Current-continuity-gated transfer

When the supervisor changes an IBR's mode, the new-mode controller states are not continued by integration; they are *re-initialised* at the current terminal operating point so that the injected current does not jump. Let the pre-switch state x^- inject $I^- = I(x^-, V)$ with $S = V\overline{I^-}$, $P = \Re S$, $Q = \Im S$. The transfer $x^- \mapsto x^+$ is

$$
x^+_{\mathrm{pl}} = x^-_{\mathrm{pl}} \ (\text{carry } i_d, i_q, V_{dc}), \qquad x^+_{\mathrm{ctrl}} = \mathrm{equilibrium\_initialize}(V, P, Q), \tag{46}
$$

with one physical carry-across of frequency: GFL→GFM sets $\omega_{\mathrm{VSG}} := \omega_{\mathrm{PLL}}$, and GFM→GFL sets the PLL integrator $\xi_{\mathrm{PLL}} := \omega_b(\omega_{\mathrm{VSG}} - 1)/k_{i,\mathrm{PLL}}$. The transfer is then *gated*: the post-switch injection $I^+ = I(x^+, V)$ must satisfy

$$\left|I^+ - I^-\right| \le \varepsilon_{\mathrm{tol}} = 10^{-10}, \tag{47}$$

otherwise the step is rejected fail-closed. Equation (47) is the bumpless-transfer guarantee: the mode may change discontinuously in *parameterisation*, but the current the network sees is continuous.

7 Switched time-domain formulation

The composite system is advanced in time by *one* coupled implicit step that the plain and the hybrid drivers share. Three questions are answered in turn, because they are the three things that make this harder than integrating a single machine: (i) the differential equations are *not* the same from one device to the next, nor even for one device across a mode change (§7.1), and the block structure that lets them still be stacked into one right-hand side is proved in §7.2; (ii) they are nonetheless solved *together*, in a single system per step, by discretising the differential part with the implicit trapezoidal rule and appending the network law (§7.3); and (iii) that system is nonlinear, so each step is closed by a damped Newton solve (§7.4). Two refinements specific to this study — the frozen coordinates (§7.5) and the mode switch that replaces integration by a direct assignment (§7.6) — follow.

7.1 Why the device ODEs are not the same

Each device owns a different differential model, so the right-hand side is heterogeneous by construction:

- the **SG** obeys the sixth-order EMF6 equations of Section 5 (swing pair, transient and sub-transient fluxes), of which five are integrated because E'_d is frozen at the singular limit (active set $[\,1, 2, 3, 5, 6\,]$);
- an **IBR in GFL** integrates its shared power stage, DC-source current and phase-locked-loop controller block — 10 states, $\mathcal{A}_{\mathrm{GFL}} = \{1{:}9, 17\}$ of (21);
- an **IBR in GFM** integrates the same power stage plus the virtual-synchronous-machine controller (no PLL) — 11 states, $\mathcal{A}_{\mathrm{GFM}} = \{1{:}3,\ 10{:}16, 17\}$;
- a **tripped** device integrates nothing ($\mathcal{A}_{\mathrm{tripped}} = \varnothing$).

The full state is the concatenation of these blocks, $x = [\,x_{\mathrm{SG}};\, x_{\mathrm{IBR}_1}; \ldots ; x_{\mathrm{IBR}_4}\,]$, and the right-hand side is the matching concatenation

$$\dot{x} \;=\; f(x, V) \;=\; \left[\, f_{\mathrm{SG}}(x_{\mathrm{SG}}, V)\,;\ f_{\mathrm{IBR}_1}(x_{\mathrm{IBR}_1}, V;\, m_1)\,;\ \ldots\,\right], \tag{48}$$

where each block f_{IBR_i} depends on that inverter's current mode $m_i \in \{\mathrm{GFL}, \mathrm{GFM}, \mathrm{tripped}\}$ and is written by the device itself. There is no single ODE that all devices share; the coupling between them is *only* through the terminal voltages V that every block reads.

7.2 Proof of the block structure of (48)

Equation (48) asserts something specific about the *model*, not about how the model is stored: that the composite vector field is the direct sum of per-device vector fields, each depending on the composite state only through its own block. This subsection states the hypotheses under which that is true, proves it, proves the converse so that the hypotheses are seen to be necessary rather than convenient, and then exhibits the one quantity in this very system that would break it.

Notation. Let there be n_d devices, device k attached to bus $b(k)$, carrying a state $x_k \in \mathbb{R}^{n_k}$, an exogenous input u_k, and a mode label m_k with values in a finite set $\mathcal{M}_k$. Put $n_x = \sum_k n_k$ and introduce the selection operators

$$S_k \in \mathbb{R}^{n_k \times n_x}, \quad x_k = S_k x, \qquad E_b \in \mathbb{R}^{1 \times n_b}, \quad V_b = E_b V, \tag{49}$$

writing e_b for the b-th standard basis column of $\mathbb{R}^{n_b}$.

Hypotheses.

(A1) Direct-sum state space. The selection operators satisfy

$$\sum_{k=1}^{n_d} S_k^\top S_k = I_{n_x}, \qquad S_j S_k^\top = 0 \quad (j \neq k), \qquad S_k S_k^\top = I_{n_k}. \tag{50}$$

This is the sentence "the composite state is the ordered concatenation of the device states" written as an operator identity instead of as index arithmetic: the second equation says the blocks are pairwise disjoint, the first says that together they exhaust the composite state.

(H1) One-port coupling. Device k exchanges energy with the rest of the system only through the current at its own terminal. Its internal dynamics therefore depend on the network only through $V_{b(k)} = E_{b(k)} V$.

(H2) Local constitutive law. There exist maps φ_k and ι_k such that

$$\dot{x}_k = \varphi_k\big(t, x_k, V_{b(k)}, u_k; m_k\big), \qquad I_k = \iota_k\big(t, x_k, V_{b(k)}, u_k; m_k\big). \tag{51}$$

(H3) Exogenous mode. On any interval between two consecutive event instants each m_k is constant, and its value is not a function of x.

Proposition 1 (Block structure)**.** *Assume* (A1)–(H3) *and fix an interval between two consecutive event instants. Then*

(i) the composite vector field is

$$f(x, V) = \sum_{k=1}^{n_d} S_k^\top \, \varphi_k\big(t,\, S_k x,\, E_{b(k)} V,\, u_k;\, m_k\big), \tag{52}$$

which written blockwise is exactly (48)*; and*

(ii) at every point at which each φ_k is differentiable in its state argument,

$$\frac{\partial f}{\partial x} = \sum_{k=1}^{n_d} S_k^\top \frac{\partial \varphi_k}{\partial x_k} S_k = \operatorname{blkdiag}\left(\frac{\partial \varphi_1}{\partial x_1}, \ldots, \frac{\partial \varphi_{n_d}}{\partial x_{n_d}}\right). \tag{53}$$

Proof. (i) By the first identity of (50), $x = \sum_k S_k^\top S_k x = \sum_k S_k^\top x_k$. Differentiating in t and substituting (51) for each $\dot{x}_k$ gives (52). Multiplying (52) on the left by S_j and using $S_j S_k^\top = \delta_{jk} I$ isolates block j as $S_j f = \varphi_j(t, S_j x, E_{b(j)} V, u_j; m_j)$, so the blocks of f are the per-device fields in device order, which is (48). Note that this part is purely algebraic: it uses no differentiability of any φ_k.

(ii) By (51) the map φ_k depends on x only through $S_k x$, so wherever the derivative exists the chain rule applied to (52) gives the first equality of (53). For the second, take $j \neq k$ and compute the (j, k) block,

$$S_j \frac{\partial f}{\partial x} S_k^\top = \sum_l \big(S_j S_l^\top\big) \frac{\partial \varphi_l}{\partial x_l} \big(S_l S_k^\top\big) = 0, \tag{54}$$

because by (50) the first factor vanishes unless $l = j$ and the third vanishes unless $l = k$, and $j \neq k$ makes at least one of them vanish in every term. The diagonal blocks are $S_k(\partial f/\partial x)S_k^\top = \partial\varphi_k/\partial x_k$. □

Corollary 1 (Where the coupling lives). *Under the same hypotheses,*

$$\frac{\partial f}{\partial V} = \sum_k S_k^\top \frac{\partial \varphi_k}{\partial V_{b(k)}} E_{b(k)},$$

so the rows of device k see exactly one bus column; and the algebraic constraint is

$$0 = g(x, V) = Y_{\text{net}} V - \sum_{k=1}^{n_d} e_{b(k)}\, \iota_k\big(t, S_k x, E_{b(k)} V, u_k; m_k\big). \tag{55}$$

Hence all interaction between devices is confined to the algebraic block, and two devices meet in it only when $b(j) = b(k)$.

Remark 1 (Solvability is a separate question). Equations (52) and (53) are statements about the map f of two independent arguments x and V. They do not require that V be determined by x; whether (55) has a unique solution is a different property, enforced at run time by the conditioning gate of Section 7.4 rather than assumed here.

Remark 2 (Why the composite field is only piecewise smooth). Part (ii) is stated pointwise because the composite field is *not* continuously differentiable everywhere: the circular limiter Π_I of Section 6 is not differentiable on the circle $|i| = I_{\max}$, and the conditional integrator $\mathcal{H}$ is not even continuous. Part (i), being algebraic, is unaffected. The finite-difference Jacobian of Section 7.4 inherits the block structure regardless of smoothness: perturbing a coordinate of x_j changes the output of block j alone, by (52), even when that perturbation carries φ_j across one of its own switching surfaces. This is one reason the step is closed with a difference quotient rather than a hand-derived derivative.

Proposition 2 (Converse). *Let $U \subseteq \mathbb{R}^{n_x}$ be open and convex and suppose $f(\cdot, V)$ is C^1 on U with $\partial f/\partial x$ block diagonal with respect to $\{S_k\}$ throughout U. Then for each k the block $S_k f$ is a function of $S_k x$ alone on U.*

Proof. Fix k and let $x, x' \in U$ satisfy $S_k x = S_k x'$. By (50) the difference $d = x' - x$ lies in $\operatorname{span}\big\{\operatorname{range} S_j^\top : j \neq k\big\}$. Convexity puts the segment $x + \tau d$, $\tau \in [0, 1]$, inside U, and along it $\frac{d}{d\tau} S_k f(x + \tau d) = S_k(\partial f/\partial x)d = 0$ because every component of d lies in a block whose $(k, \cdot)$ derivative block vanishes. Hence $S_k f(x') = S_k f(x)$. □

Convexity, or a local statement, is needed rather than mere connectedness: on a connected but non-convex U a vanishing directional derivative gives constancy only along each connected component of the corresponding slice.

Remark 3 (The hypotheses have content: a system-wide signal breaks them). Proposition 2 shows the block structure holds *if and only if* the device dynamics are local, so (H2) cannot be dropped. This system in fact contains a quantity that would violate it. The centre-of-inertia frequency is the inertia-weighted mean

$$f_{\mathrm{coi}}(t) = \frac{\sum_{i\in\mathcal{V}(t)} H_i f_i(t)}{\sum_{i\in\mathcal{V}(t)} H_i}, \qquad \mathcal{V}(t) = \{\, i : \ i \text{ online with } H_i > 0 \,\}, \tag{56}$$

in which $\mathcal{V}(t)$ comprises the synchronous machine and every grid-forming converter, a grid-following converter having no inertia to contribute; and $f_{\mathrm{SG}} = f_0(1+\omega)$ and $f_{\mathrm{GFM},i} = f_0(1+\omega_{\mathrm{VSG},i})$ are *states*. Consider the hypothetical modification — *not* the model of this report — in which one converter's swing acquires a system-wide droop term, $M\dot{\omega}_{\mathrm{VSG},k} = \kappa u_1 - P_k - D_v(\omega_{\mathrm{VSG},k} - 1) - D_g\big(f_{\mathrm{coi}}(x) - f_0\big)$. That map is not of the form (51), and $\partial(S_k f)/\partial x_j \neq 0$ for every $j \in \mathcal{V}(t)$, so (53) fails and the Jacobian acquires a dense block row. The actual model escapes this because f_{coi} enters only the severity index of Section 4, which acts on the discrete label m_k and therefore through (H3); and because $\mathcal{V}(t)$ is itself determined by the mode vector, which is a second reason the quantity belongs to the switching layer and not to the vector field.

Remark 4 (The object is a switched system). Formally the model is a triple: a finite family of vector fields $\{f^{(m)}\}_{m\in\mathcal{M}_1\times\cdots\times\mathcal{M}_{n_d}}$, a switching signal $t \mapsto m(t)$ that is piecewise constant by (H3), and a reset map applied at the switching instants (Section 7.6). Equation (48) is accordingly a statement about each interval between consecutive events, not about the trajectory as a whole.

Remark 5 (Two operators, not one). The layout operators S_k of (49) are constant: a device's stored state count does not change with its mode. What changes with the mode is the *active* set, carried by a separate projection $P_{\mathcal{A}}$ (Section 7.5). Writing $\mathcal{A}_k$ for device k's local active set and $o_k = \sum_{j<k} n_j$ for its offset, the composite active set is the disjoint union

$$\mathcal{A} = \bigsqcup_{k=1}^{n_d} \big(o_k + \mathcal{A}_k\big), \tag{57}$$

disjoint precisely *because* the S_k are mutually orthogonal by (50). A tripped device contributes $\mathcal{A}_k = \varnothing$, and a singular-limit coordinate such as the E'_d of Section 5.3 is removed from $\mathcal{A}_k$ by its own device.

Hypotheses (A1) and (H1)–(H3) are properties of the device models set out in Sections 5 and 6, not of any particular implementation of them; the provenance of those models is given in Section 2.

7.3 One coupled DAE, one implicit trapezoidal step

The heterogeneous ODE (48) is coupled to the all-KCL network law (1) to form the composite differential–algebraic system

$$\dot{x} = f(x, V), \qquad 0 = g(x, V) = Y_{\mathrm{net}}V - I_{\mathrm{bus}}(x, V). \tag{58}$$

It is advanced from (x_0, V_0) to (x_1, V_1) over a step h by discretising the differential rows with the implicit trapezoidal rule and carrying the algebraic rows exactly. With $f_0 = f(x_0, V_0)$,

$f_1 = f(x_1, V_1)$ and the active set $\mathcal{A} \subseteq \{1, \ldots, n_x\}$ (the union of the per-device active sets above), the step solves

$$\underbrace{\left[x_1 - x_0 - \tfrac{h}{2}(f_0 + f_1)\right]_{\mathcal{A}} = 0}_{\text{active trapezoidal rows}}, \qquad \underbrace{g(x_1, V_1) = 0}_{\text{all KCL rows}}. \tag{59}$$

Because f_1 and g are evaluated at the unknown end-of-step state, this is an *implicit* (nonlinear, coupled) system, not an explicit update: the states of every device and the voltages of every bus are unknowns of one and the same system, so the machine and all four inverters are solved simultaneously and are never sub-iterated device by device. For an active state i the first block is exactly the trapezoidal rule $x_{1,i} = x_{0,i} + \frac{h}{2}(f_{0,i} + f_{1,i})$.

7.4 How the step is solved: the damped Newton system

Stacking the unknowns as $z = [\, x_1|_{\mathcal{A}};\, V_1 \,]$, the two blocks of (59) become a single nonlinear residual

$$r(z) = \begin{bmatrix} \left[x_1 - x_0 - \frac{h}{2}(f_0 + f_1)\right]_{\mathcal{A}} \\ g(x_1, V_1) \end{bmatrix} = 0, \tag{60}$$

which is driven to zero by a damped Newton iteration. This is the single Newton owner reused by the equilibrium solve, the trapezoidal step, and the event solve; no device supplies its own solver. One iteration is:

1. form the Jacobian $J = \partial r / \partial z$ by *forward finite differences* (column j is $[r(z + \varepsilon e_j) - r(z)]/\varepsilon$ with $\varepsilon = 3 \times 10^{-6}$); no analytic Jacobian is assembled, so a device may change its equations without hand-derived derivatives, and by Remark 2 the difference quotient stays well-defined and block structured where the limiter and anti-windup surfaces make the field non-differentiable;
2. test conditioning: if $\mathrm{rcond}(J) < \epsilon_{\mathrm{mach}}$ the step is rejected fail-closed rather than solved through a singular system;
3. take the Newton direction $\Delta z = -(J \setminus r)$ with the audited MATLAB backslash — the only linear solve in the loop;
4. damp it with a backtracking line search: try $z + \alpha\, \Delta z$ at $\alpha = 1$ and halve α (up to 20 times) until $\|r\|_\infty$ strictly decreases; if no α decreases the residual the step fails closed at the last accepted iterate.

The iteration is declared converged when $\|r(z)\|_\infty < \mathrm{tol} = 10^{-8}$, and at most 50 iterations are taken. Nothing in the loop is tuned to the case: the tolerance, the finite-difference increment, the conditioning gate and the line-search contract are fixed constants of the solver.

7.5 Frozen coordinates: the Π-projection

Every index *not* in $\mathcal{A}$ — the frozen set $\mathcal{F} = \{1, \ldots, n_x\} \setminus \mathcal{A}$ — is removed from the Newton system and held exactly:

$$x_{1,i} = x_{0,i} \quad \text{for all } i \in \mathcal{F}. \tag{61}$$

The solver begins from $x_1 \leftarrow x_0$ and never writes a frozen row, and a drift check re-verifies $x_1|_{\mathcal{F}} = x_0|_{\mathcal{F}}$ after the step; any change of a frozen coordinate is a hard error. Two distinct things live in

$\mathcal{F}$: the singular-limit flux E'_d of Section 5.3 (frozen for the whole run), and the entire controller block that a given IBR is *not* currently using (frozen for as long as that mode is inactive). The map Π that selects $\mathcal{A}$ is the same runtime authority (`ts_dynamic_state_indices`) at every one of its call sites, so the reported dimension $n_x(t)$ cannot disagree with what the integrator actually solved.

7.6 The switch: from integration to assignment

A mode change is landed *exactly* on a step boundary (the step is subdivided so the event time is a grid point). At that boundary the state is not advanced by (59); it is replaced by the discrete transfer of Section 6.6:

$$x_1 \;=\; \mathcal{T}(x_0, V) \quad \text{(a re-initialisation), not} \;\; x_1 = x_0 + \tfrac{h}{2}(f_0 + f_1). \tag{62}$$

At the mode boundary the controller coordinates jump to the transferred values $\mathcal{T}(x_0, V)$ chosen to satisfy the current-continuity gate (47), and integration resumes on the *new* active set on the very next step. Two important consequences follow for this study:

- The **SG trip** is *not* such an assignment. The SG keeps the same active set $[\,1, 2, 3, 5, 6\,]$; its trajectory is continuous across $t = 20$ s (the coast of Section 5). Only the right-hand side branch changes (connected stator → open circuit).
- The **IBR GFL→GFM** changes *are* such assignments: each IBR's active dimension steps $9 \to 10$ and its controller states are re-initialised by (46). The composite dynamic dimension therefore changes because of the inverters, not the generator. In the delivered run these assignments land at the supervisor's own commitment instants (22.089, 51.340, 53.379, 148.281 s), none of which coincides with a scheduled disturbance time.

8 Power-flow results

Table 2 is the solved SG-online power flow (the reference operating point from which $V_i^\star$ in (2) is taken); Table 3 the corresponding branch flows and losses. The reference bus supplies $P_g = 1.3319$ p.u., $Q_g = -0.1957$ p.u.; the four IBR buses (2, 3, 6, 8) inject at their scheduled set-points while every load bus meets its P_L, Q_L.

Table 2: Solved bus power flow at the SG-online operating point (per unit on the system base; angles in degrees).

Bus	Type	$\lvert V\rvert$	θ (deg)	P_g	Q_g	P_L	Q_L	Q_{min}	Q_{max}	P_{max}
1	REF	1.0000	0.0000	1.3319	-0.1957	0.0000	0.0000	0.00	0.10	3.32
2	GFL	0.9914	-3.2585	0.3172	0.1767	0.2170	0.1270	-0.40	0.50	1.40
3	GFL	0.9667	-8.6921	0.2884	0.1607	0.9420	0.1900	0.00	0.40	1.00
4	PQ	0.9832	-6.7064	0.0000	0.0000	0.4780	-0.0390	–	–	–
5	PQ	0.9858	-5.5221	0.0000	0.0000	0.0760	0.0160	–	–	–
6	GFL	1.0575	-6.7136	0.4443	0.2476	0.1120	0.0750	-0.06	0.24	1.00
7	PQ	1.0253	-6.9205	0.0000	0.0000	0.0000	0.0000	–	–	–
8	GFL	1.0494	-4.4090	0.2677	0.1491	0.0000	0.0000	-0.06	0.24	1.00
9	PQ	1.0219	-8.6416	0.0000	0.0000	0.2950	0.1660	–	–	–
10	PQ	1.0202	-8.5886	0.0000	0.0000	0.0900	0.0580	–	–	–
11	PQ	1.0347	-7.7782	0.0000	0.0000	0.0350	0.0180	–	–	–
12	PQ	1.0411	-7.6862	0.0000	0.0000	0.0610	0.0160	–	–	–
13	PQ	1.0342	-7.8345	0.0000	0.0000	0.1350	0.0580	–	–	–
14	PQ	1.0087	-9.3374	0.0000	0.0000	0.1490	0.0500	–	–	–

Table 3: Branch power flows and losses at the SG-online operating point.

Line	From	To	P_{from}	Q_{from}	P_{loss}	Q_{loss}
1	1	2	0.910918	-0.153112	0.016392	-0.002303
2	1	5	0.420987	-0.042553	0.009593	-0.008907
3	2	3	0.466697	0.013221	0.010470	0.002115
4	2	4	0.316733	-0.064476	0.006065	-0.014738
5	2	5	0.211276	-0.049810	0.002649	-0.025732
6	3	4	-0.197333	-0.018164	0.002803	-0.005014
7	4	5	-0.448842	0.084724	0.002882	0.009089
8	4	7	0.018419	-0.096315	0.000000	0.001990
9	4	9	0.062955	-0.012297	0.000000	0.002223
10	5	6	0.092297	0.001911	0.000000	0.001920
11	6	11	0.130821	0.059907	0.001758	0.003682
12	6	12	0.086088	0.027130	0.000895	0.001864
13	6	13	0.207667	0.085524	0.002983	0.005875
14	7	8	-0.267650	-0.134127	0.000000	0.015018
15	7	9	0.286069	0.035823	0.000000	0.008698
16	9	10	-0.003258	0.021664	0.000015	0.000039
17	9	14	0.057282	0.023368	0.000466	0.000991
18	10	11	-0.093273	-0.036375	0.000790	0.001849
19	12	13	0.024192	0.009266	0.000137	0.000124
20	13	14	0.093739	0.030791	0.001556	0.003168

9 Small-signal spectrum

The modal table below lists the small-signal eigenvalues at a representative post-disturbance configuration (one IBR grid-forming, three grid-following): index, the real and imaginary parts of the eigenvalue λ in separate columns, the modal frequency f, the damping ratio ζ, and the dominant state — the single coordinate with the largest normalised participation. Conjugate pairs are printed once, so a tabulated imaginary part is written ± and stands for both members of the pair; a purely real mode prints 0 there and $\zeta = 1$. Rows whose two leading participations are too close to rank carry a dagger and are counted in the table's footnote, so a named state is never read as a sole attribution when it is not one. Participation is a reporting diagnostic and enters no gate. It is the generated table itself, not a re-typeset copy.

No.	$\Re\lambda$ (s^{-1})	$\Im\lambda$ (s^{-1})	f (Hz)	ζ	Dominant state
1	−2075.371442	0	0	1	IBR6:i_d
2	−1459.095967	0	0	1	IBR8:i_q
3	−1408.957478	0	0	1	IBR3:i_d
4†	−1288.036461	0	0	1	IBR6:i_q
5	−1252.898506	0	0	1	IBR8:i_d
6	−952.908217	±403.267239	64.181975	0.920928	IBR2:i_q
7	−900.503570	0	0	1	IBR2:i_d
8	−249.267967	0	0	1	IBR2:ω_{VSG}^{GFM}
9	−159.268727	0	0	1	IBR2:E^{GFM}
10†	−98.406155	±98.380337	15.657717	0.707200	IBR2:I_{dc}
11†	−96.455579	±96.325245	15.330639	0.707585	IBR8:V_{dc}
12†	−96.346517	±96.207876	15.311959	0.707616	IBR3:V_{dc}
13†	−95.555521	±95.348575	15.175197	0.707873	IBR6:V_{dc}
14	−13.544228	±0.055071	8.7648×10^{-3}	0.999992	IBR2:ξ_{Id}^{GFM}
15	−13.507642	0	0	1	IBR2:ξ_{Id}^{GFM}
16†	−13.492443	0	0	1	IBR8:ξ_{Id}^{GFL}
17	−13.488369	0	0	1	IBR6:ξ_{Iq}^{GFL}
18	−13.475757	0	0	1	IBR3:ξ_{Id}^{GFL}
19	−13.471825	0	0	1	IBR8:ξ_{Iq}^{GFL}
20	−13.433124	0	0	1	IBR6:ξ_{Id}^{GFL}
21	−2.623074	0	0	1	IBR2:ξ_{Vq}^{GFM}
22†	−1.843600	0	0	1	IBR8:ξ_{P}^{GFL}
23	−1.491476	0	0	1	IBR8:ξ_{Q}^{GFL}
24	−1.434890	0	0	1	IBR3:ξ_{P}^{GFL}
25	−1.364330	0	0	1	IBR6:ξ_{P}^{GFL}
26	−1.326920	0	0	1	IBR3:ξ_{Q}^{GFL}
27	−1.185526	±2.980707	0.474394	0.369574	IBR2:ξ_{Vd}^{GFM}
28†	−0.588912	±2.171749	0.345645	0.261717	IBR6:δ_{PLL}^{GFL}
29†	−0.566771	±2.132137	0.339340	0.256901	IBR3:δ_{PLL}^{GFL}
30	−0.328014	±0.448860	0.071438	0.590017	IBR2:ξ_{Vd}^{GFM}

† 9 of the 30 rows carry this mark. Their two leading participations differ by less than 0.02, so the named state is not a sole attribution there.

The spectrum is entirely in the left half-plane at this configuration. This is a small-signal statement about one linearisation point; it is not, and must not be read as, a stability certificate for the full switched trajectory.

The DC-link modes are a check on the model, not a free observation. Each converter contributes two DC coordinates, V_{dc} and the source current I_{dc}, and Section 6.3 predicts their joint behaviour *before* the spectrum is computed. Equation (28) is a 2×2 block whose maximally-flat design target is $\zeta = 1/\sqrt{2}$, so the table must show, per converter, one complex conjugate pair with damping ratio 0.707 and a modal frequency near 15 Hz — and the four pairs must differ from one another, ordered by the power each link carries, because the loading enters A_{dc} through $a = P^0_{ac}/(C(V^0_{dc})^2)$. The table satisfies every part of that: the four pairs are -98.41 ± 98.38j, -96.46 ± 96.33j, -96.35 ± 96.21j and -95.56 ± 95.35j, at 15.18–15.66 Hz with $\zeta = 0.7072$–0.7079. Nothing was fitted to obtain that agreement: τ_s was derived from the filter criterion and the spectrum was computed afterwards.

Reading the DC participation, and what it does and does not say. Each of those pairs carries a participation of exactly 0.500 on that converter's V_{dc} and 0.500 on its own I_{dc}, with every other coordinate at 0.000. Both halves of that sentence are structural and were measured on the assembled system rather than assumed.

The *equal split* is the symmetry of a 2×2 block with a conjugate pair: the 8×8 DC submatrix of the assembled model is exactly block-diagonal by converter, the measured largest entry outside those four 2×2 blocks being 0, so each pair lives in a two-dimensional invariant subspace and the two coordinates in it participate equally.

The *zeros elsewhere* are the one-way coupling, which the source state does not change. Perturbing either DC coordinate changes no AC state derivative at all,

$$\max_{i \in \mathrm{AC}} \max_{j \in \{V_{dc}, I_{dc}\}} \left| \frac{\partial f_i}{\partial x_j} \right| = 0 \quad \text{exactly, on the assembled } 40 \times 40 \text{ matrix,} \tag{63}$$

while the DC rows do read the AC states, with a largest entry of 33.4. So the DC family is a closed sub-system driven by the AC side, and no DC participation ever reaches i_d or i_q. A reader should take 0.500/0.500 as "this mode is the DC circuit of one converter, shared between its two coordinates", not as "the DC circuit is coupled into the network".

The modelling limitation behind (63). That one-way coupling is a property of this model, not of a real converter. The inner loop forms its commanded voltage v_{cd}, v_{cq} from the current error alone; nothing ties that command to the DC voltage available to synthesise it. A physical two-level converter can only produce roughly $m\, V_{dc}/2$ per phase, so a sagging link clips the AC command, the coupling becomes two-way, and the DC participation would spread beyond the two DC coordinates. Adding that modulation limit is a change to the AC equations and is not made here. Until it is, the DC circuit of this study responds to the AC side but cannot act back on it, and no claim in this report depends on it doing so.

Three revisions of the same coordinate. It is worth recording what changed and what did not, because the participation column has carried three different readings. The first revision printed *four identical* rows at $\lambda = -10.000000$ with participation 1.000: that was an ideal source, whose exact power feed-forward decoupled the DC row from everything, fixed its eigenvalue at $-1/T_{dc}$ with multiplicity four, and held V_{dc} identically constant. The second printed *four distinct real* roots, still with participation 1.000, because a resistive source gives each converter one DC coordinate and its own loading-dependent eigenvalue. The present revision prints *four distinct pairs* with participation split 0.500/0.500, because the source circuit has an inductance and therefore a second coordinate. Only the first of those three was an idealisation; the second and third differ by whether τ_s is zero, and Section 6.3 shows the second is the limit of the third.

The same analysis over every admissible grid-forming set. The single configuration above is one row of a larger enumeration. With the SG offline, every subset of the four converters that could hold the voltage-forming duty is solved for *its own* equilibrium and linearised on the full-KCL composite system. There are 15 such candidates and only 7 of them are admissible, with the best damping margin at $n = 2$ units ($IBR3 + IBR6$, $\Omega = -0.56814$) rather than at the largest set ($\Omega = -0.48291$). Two facts follow and neither is a matter of taste. The admissible sets are *not* monotone in the number of grid-forming units, so the table has to be *computed* and cannot be guessed from a count. And a set can be admissible here and still not be reachable by a trajectory, which is why this screen is a necessary and not a sufficient condition. The per-configuration table and figure are held out of this revision with the control-policy comparison they belong to.

10 Time-Domain Simulation

Figure 4 places the six-period timeline directly above every result figure of this section, so that an event instant can be matched against a curve without turning back a page.

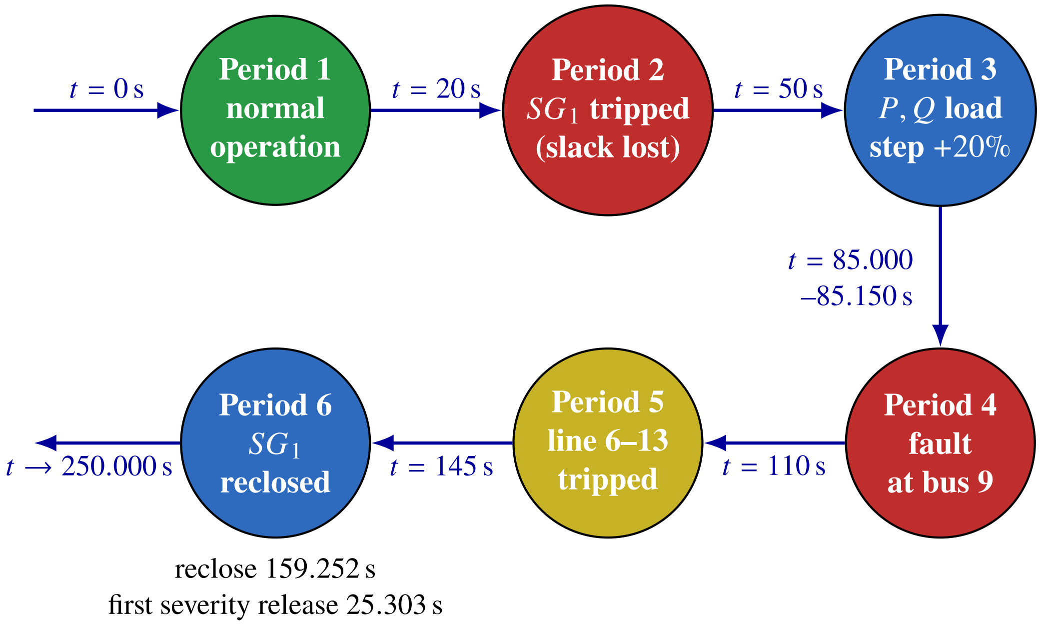

Figure 4: Six-period disturbance chronology. Green: undisturbed. Red: a contingency that removes a source or faults a bus. Yellow: a topology change. Blue: a restorative action. Scheduled instants (0, 20, 50, 85.000–85.150, 110, 145 s) are case-defined; the reclose and first severity-release instants are read from the accepted event log.

Table 5 collects the summary scalars of the run that draws every figure in this section. Each is an accepted value of the 250.000 s adaptive arm, read through a generated macro rather than typed into the prose. The voltage and frequency extremes are *reported results* of a deliberately severe event sequence, not acceptance criteria, and no model was adjusted after they were seen.

Table 5: Accepted summary scalars of the 250.000 s run that draws the figures of this section.

Quantity	Value	Note
Last accepted time	250.000 s	full horizon; no truncation
Reclose outcome	SUCCESS	read from the accepted event log
Network voltage range	0.036855–1.523199 p.u.	transient extrema, not ratings
COI frequency range	58.839081–61.138547 Hz	fault and switching window
Terminal COI frequency	60.000001 Hz	at $t = 250.000$ s
Terminal voltage range	0.9667–1.0575 p.u.	at $t = 250.000$ s
Largest accepted step residual	9.9721×10^{-9}	below the 10^{-8} gate at every step
Deepest step subdivision	0 levels	bisection mechanism
Domain-rejected trial steps	1027	no gate was relaxed to pass
Controller-rejected steps	161	adaptive stepper
Accepted samples	3679	
SG reclose instant	159.252 s	outcome, not a schedule
Closing mismatch	$\Delta V = 0.008785$ p.u. $\Delta f = 1.8805 \times 10^{-8}$ pu $\Delta\theta = 5.015017°$	within the existing synchronism frame
Mode augmentations (4)	22.089, 51.340, 53.379, 148.281 s	each passed a transition certificate
Mode releases (3)	25.303, 146.278, 152.402 s	one device at a time
Maximum simultaneous GFM units	4	reached in stages, not at once
Post-reclose mode return (1)	at 173.1271 s	the last grid-forming unit handed back
Post-reclose reselection status	SUCCESS	authenticated against the SG-online table

Figure 5 reads top to bottom as cause and then effect. Panel (a) is the severity index S of Section 4 with both thresholds drawn, one trace per converter because J_V is each converter's own terminal deviation; it is the only quantity on the page that *causes* anything. Immediately below it are two supervisory strips: each IBR's grid-forming/grid-following mode, and the *reference owner* — the single device that holds the island angle reference at each instant. Panels (b) and (c) are the per-converter active and reactive power over the full horizon 0–250.000 s. Mode and reference ownership are distinct contracts: several IBRs can be grid-forming at once, but only one of them owns the reference. Here the SG owns it up to $t = 20$ s and again after the reclose; in between, IBR_1 (bus 2) is the reference leader.

The cycle closes: every converter returns to grid-following. Once the machine is back and the C_1 command ramp has finished, the supervisor revisits the mode map one last time. At $t = 173.1271$ s — the first sample after the ramp completes, $173.1271 = 159.252 + 13.8752$ — it commits the one remaining grid-forming unit, IBR_1, back to grid-following, authenticated as an exact one-step release against the SG-online table: the all-grid-following configuration is admissible on the declared damping criterion with worst $\zeta = 0.156$ against the 0.05 requirement, and the machine owns the angle reference from the reclose onwards, so no grid-forming source is needed. The terminal state is all-grid-following with the machine as sole reference owner — the same configuration the coordinated hand-back reaches by its own route. The full cycle is therefore closed: the machine trips, one converter takes the voltage-forming duty, support units are augmented on severity and released when it subsides, the machine returns, and every converter goes back to grid-following.

The point of this result set is in the mode strip: the supervisor does *not* raise all four converters together. It augments in 4 stages, at 22.089, 51.340, 53.379 and 148.281 s, until it holds 4 grid-forming units, then releases 3 times after the reclose, before the final one-device return above. *Pinning* those same 4 units from the trip onwards instead destroys the run at $t = 25.488$ s with `adaptiveDtMin`, even though that set passes the linear admissibility check. The order in which the set is reached therefore matters as much as the set itself.

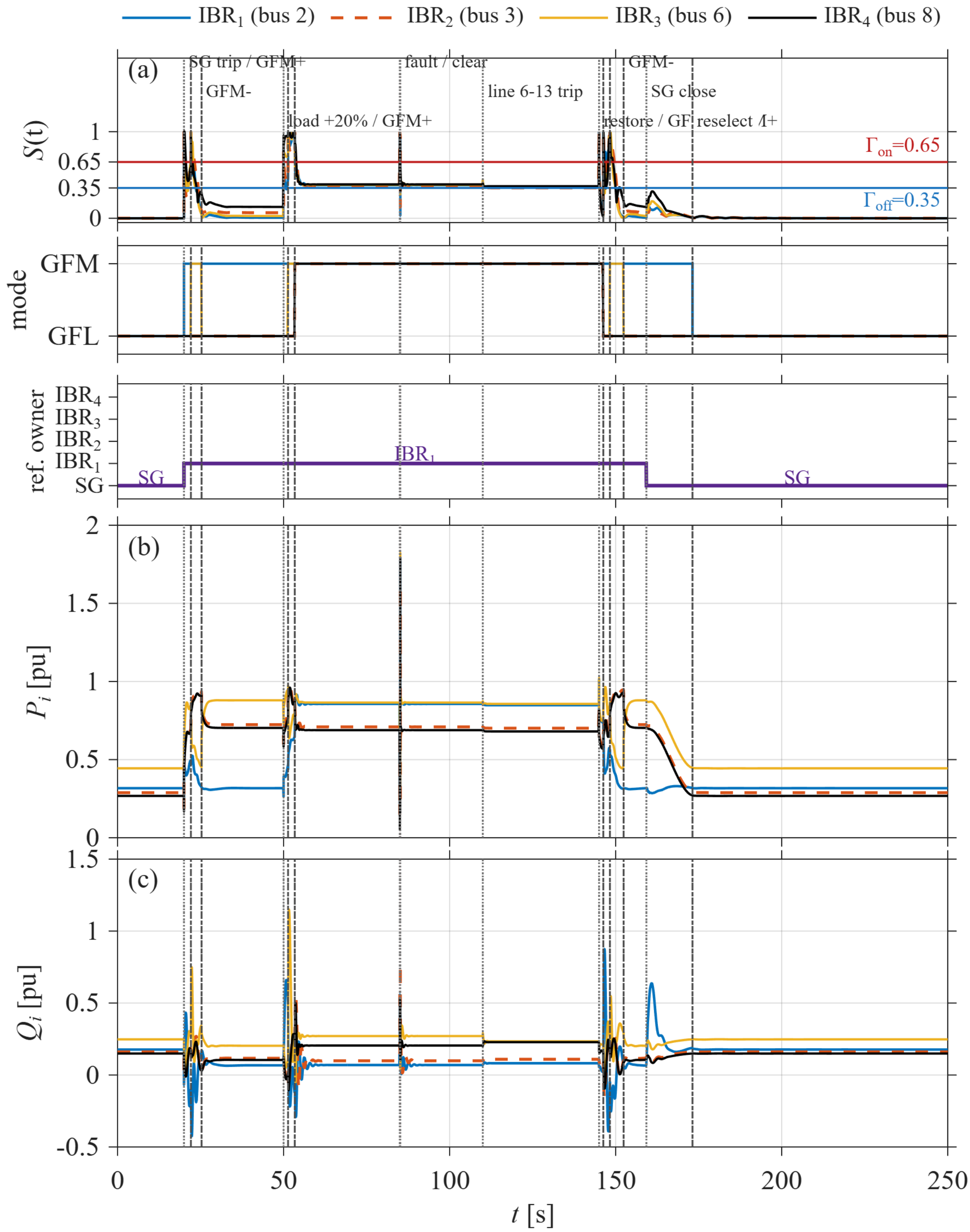

Figure 5: (a) Severity index S per converter with both switching thresholds, $\Gamma_{\text{on}} = 0.65$ and $\Gamma_{\text{off}} = 0.35$. Second strip: grid-forming or grid-following mode of each IBR. Third strip: the single device holding the island angle reference. (b) Per-IBR active power P_i. (c) Per-IBR reactive power Q_i, both per unit. Dotted lines are scheduled disturbances and dash–dotted lines the supervisor's own commitments; near-coincident instants share one label. Raw accepted samples over the full 250.000 s horizon.

Figure 6 widens the same trajectory into the eight electrical quantities that the device reconstruction publishes. Panel (h) is a whole-network scalar and therefore has one trace by definition;

every other panel carries the four converters and, where a synchronous counterpart exists, the SG.

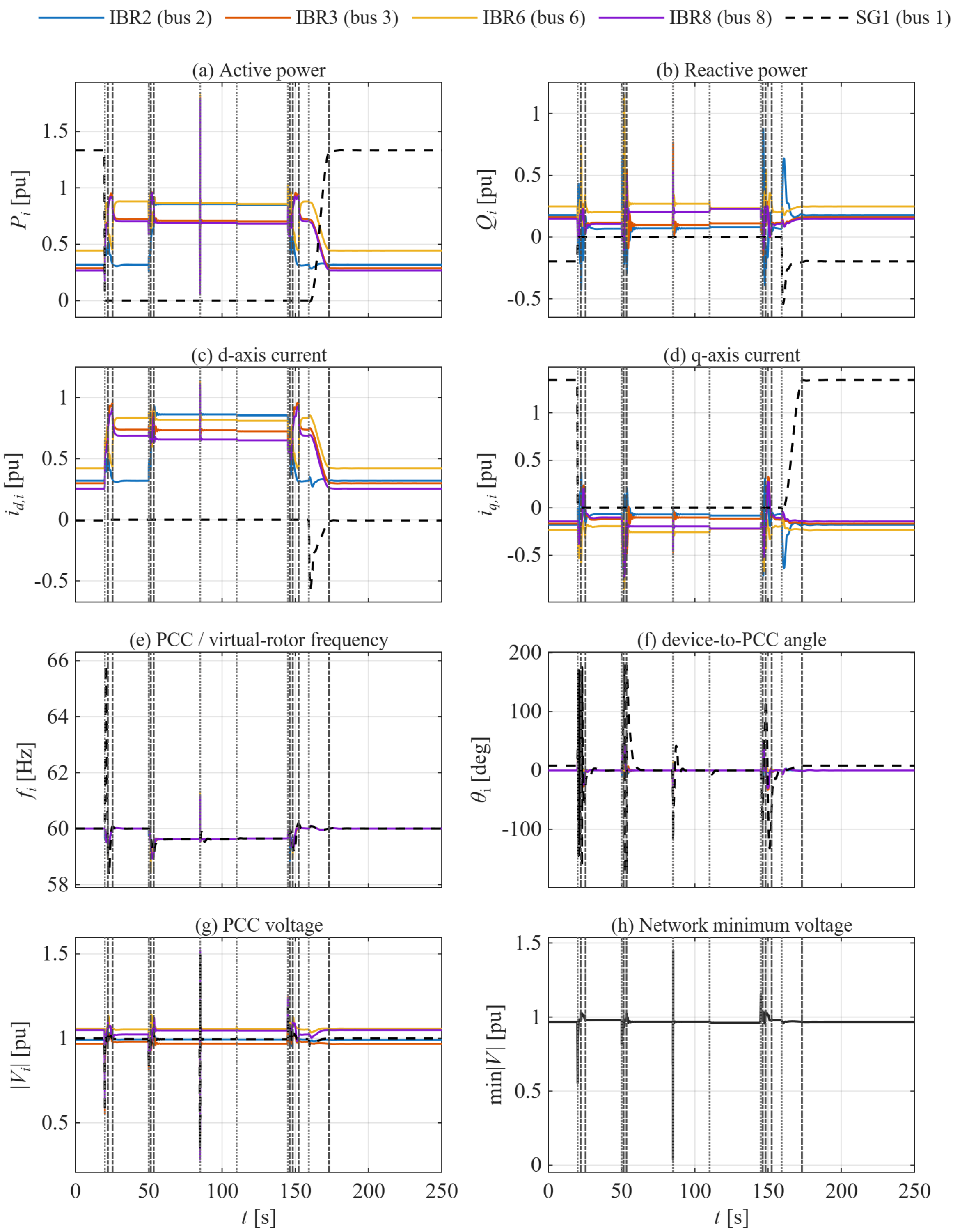

Figure 6: Eight-panel electrical response over the full 250.000 s horizon. (a) active power; (b) reactive power; (c) d-axis current; (d) q-axis current; (e) PCC or virtual-rotor frequency; (f) device-to-PCC angle; (g) PCC voltage; (h) network minimum voltage. The SG is dashed black wherever a synchronous counterpart exists; panel (h) is a system scalar and has a single trace by definition. Vertical lines mark the same instants as Figure 5, unlabelled here because each panel is half a page wide. Raw accepted samples.

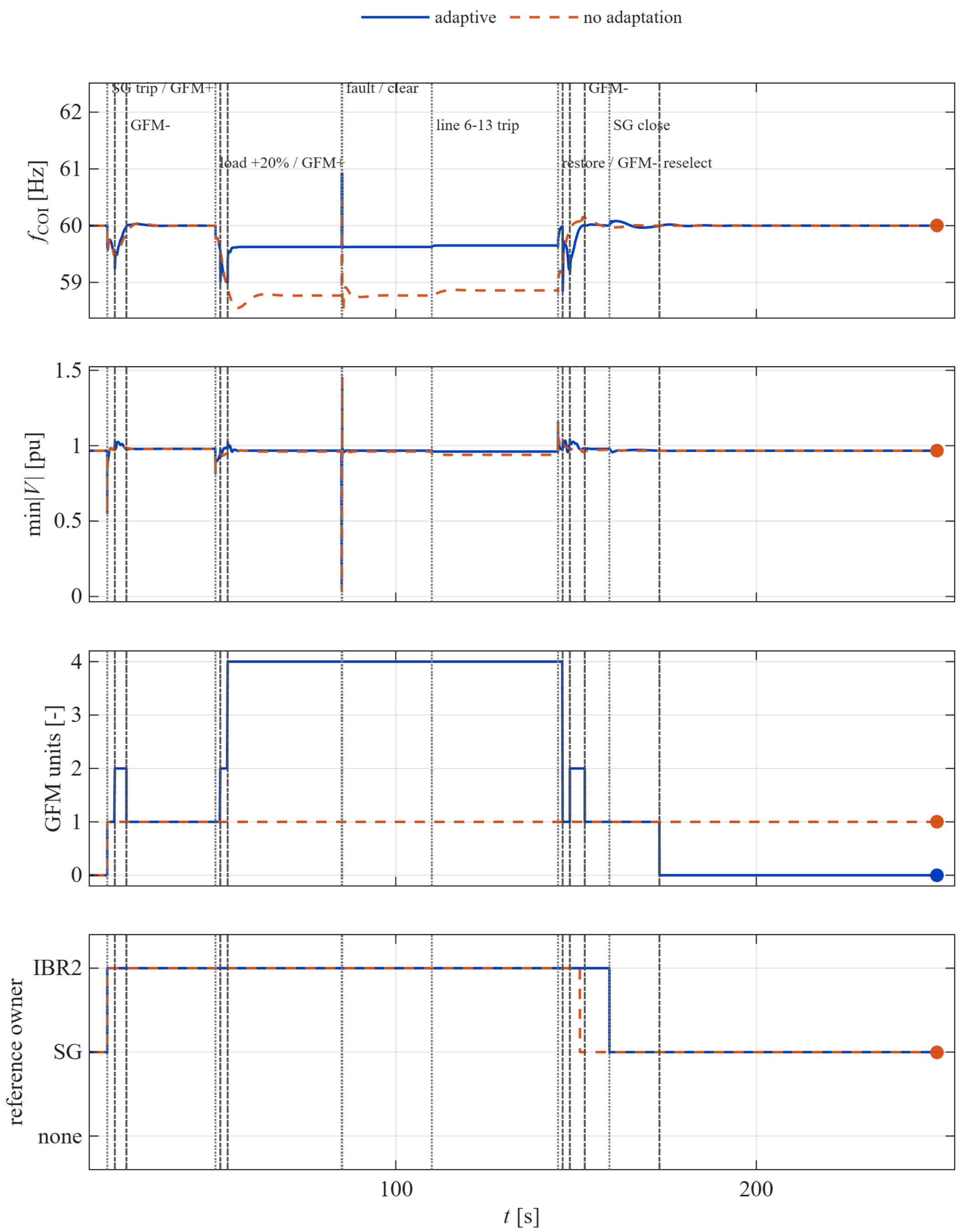

Figure 7: Switching against not adapting, over the whole horizon from five seconds before the machine trip. Solid: the delivered adaptive policy. Dashed: the no-adaptation control, which commits one grid-forming unit — the best single-unit configuration of the admissibility table — at the trip and never revisits that choice. Top: COI frequency, where the control rides 0.4–1.1 Hz lower through every islanded disturbance and only recovers once the machine returns; the two traces coincide again after the hand-back. Second: network minimum voltage. Third: grid-forming unit count — the control holds exactly one throughout, the adaptive policy raises four under stress and returns to zero after the mode return. Bottom: reference owner, which both policies hand back to the machine, after the C_1 ramp in both policies. The control is not a collapse: it runs to the horizon. Its cost is a persistent frequency deficit — a peak of 1.4978 Hz from nominal at the fault and 1.2325 Hz sustained after the line trip, against 1.1385 Hz and 0.3764 Hz for the adaptive policy — which is the physical price of carrying the island on the minimum grid-forming capacity with no droop support added under stress.

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